

The Flight of ISEE-3/ICE: Origins, Mission History, and a Legacy¹

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Abstract

A brief history of the ISEE-3/ICE mission is presented. ISEE-3/ICE was the first spacecraft to use a “halo orbit” and lunar gravity-assist maneuvers to achieve its mission objectives. Using these innovative trajectory techniques, ISEE-3/ICE was able to carry out a number of important scientific mission tasks including: the first continuous measurements of solar-wind conditions upstream from the Earth; the first exploration of the Earth’s magnetotail between 80 and 240 Earth radii; and the first encounter with a comet. The account of the ISEE-3/ICE flight history is augmented by discussions of the political and programmatic hurdles that had to be overcome to gain approval to redirect ISEE-3/ICE to the distant geotail region, and then on to a comet. Key documents related to the approval of these extended missions are reproduced in the appendices. Finally, it is shown that, by successfully demonstrating the usefulness and practicality of halo orbits and lunar gravity-assist trajectories, ISEE-3/ICE has paved the way for numerous follow-on missions such as Hiten, Geotail, Wind, SOHO, ACE, MAP, Genesis, Triana, FIRST, Planck, and NGST.

Introduction

On August 12, 1978, a spacecraft called the International Sun-Earth Explorer-3 (ISEE-3) was launched towards the Sun-Earth L1 libration point. This was the beginning of a space odyssey that included four orbits around the Sun-Earth L1 point, lengthy traverses of the Earth’s magnetotail, five lunar swingby maneuvers, and a pass through the tail of a comet. Now, some twenty years later, ISEE-3 [renamed International Cometary Explorer (ICE) in 1983] is located in a heliocentric orbit about 2 AU from the Earth on the other side of the Sun. ISEE-3/ICE will return to the Earth’s vicinity in August 2014.

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During its meandering journey in the Sun-Earth-Moon system, ISEE-3/ICE has racked up an impressive list of space “firsts” including the following:

- First spacecraft stationed in a libration-point orbit.
- First continuous monitor of solar-wind conditions upstream from the Earth.
- First real-time warning system for geomagnetic storms.
- First use of multiple lunar swingbys for orbital control in the Earth-Moon System.
- First exploration of Earth’s magnetotail between 80 and 240 Earth radii.
- First use of lunar gravity-assist maneuver to “launch” spacecraft towards a body outside the Earth-Moon system.
- First encounter with a comet.

As shown in this paper, the distinguishing feature of the ISEE-3/ICE mission is its unique trajectory design. ISEE-3/ICE was the first spacecraft to use two fundamental trajectory concepts that are enabling factors in an increasing number of scientific space missions. The two concepts are known as “Halo Orbits” and “Double Lunar-Swingby” (DLS) trajectories. The successful flight of ISEE-3/ICE demonstrated both the utility and the practicality of halo orbits and DLS trajectories. The primary purpose of this paper is to present an historical account of the ISEE-3/ICE mission, and its importance to future libration-point and lunar gravity-assist missions.

Earth-Moon Libration Points and the Genesis of the Halo-Orbit Concept

In 1772, the eminent French mathematician, J. L. Lagrange, showed that there are five positions of equilibrium in a rotating two-body gravity field. Three of these “libration points” (sometimes called Lagrangian points) are situated on a line joining the two attracting bodies, and the other two form equilateral triangles with these bodies. All five libration points lie in the orbit plane of the two primary bodies. A sketch of the libration points in the Earth-Moon System is shown in Fig. 1.

It is well known that the collinear points are unstable, and that the equilateral-triangle points are only quasi-stable. Therefore, if for some reason, we would want to place a spacecraft at a libration point, some form of stationkeeping control would be required (e.g., thrust control). In a little-known, but very important paper, G. Colombo showed that the ΔV cost for maintaining a satellite in the vicinity of a collinear point was mainly a function of the accuracy of its orbit determination, and could be as small as 10 m/sec per year [1].

Although the existence of libration points in the Earth-Moon System was well known to the early pioneers of space flight, no one mentioned potential applications of these special locations until 1950 when the space-travel visionary Arthur C. Clarke suggested that the translunar libration point, L2, would be an ideal site for TV and radio coverage to colonies on the far side of the Moon [2, 3]. Of course, a communication satellite stationed at the L2 point would be permanently hidden from the Earth, and would need another satellite to complete the Earth to lunar far-side link. An early concept for a continuous lunar farside communications system using comsats at both the L2 and L4 libration points is illustrated in Fig. 2 [4].

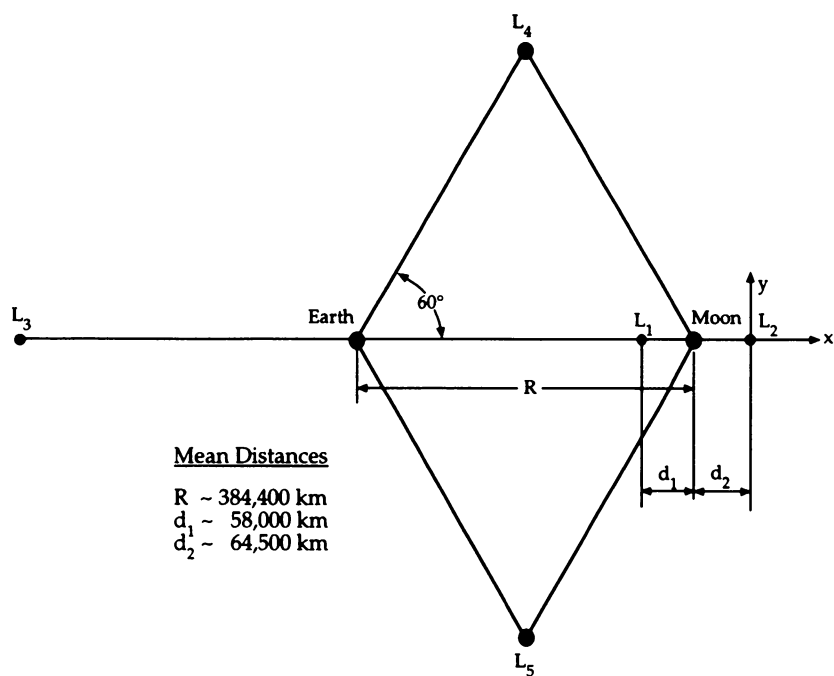


FIG. 1. Libration Points in the Earth-Moon System (*xy*-plane lies in Moon's Orbital Plane).

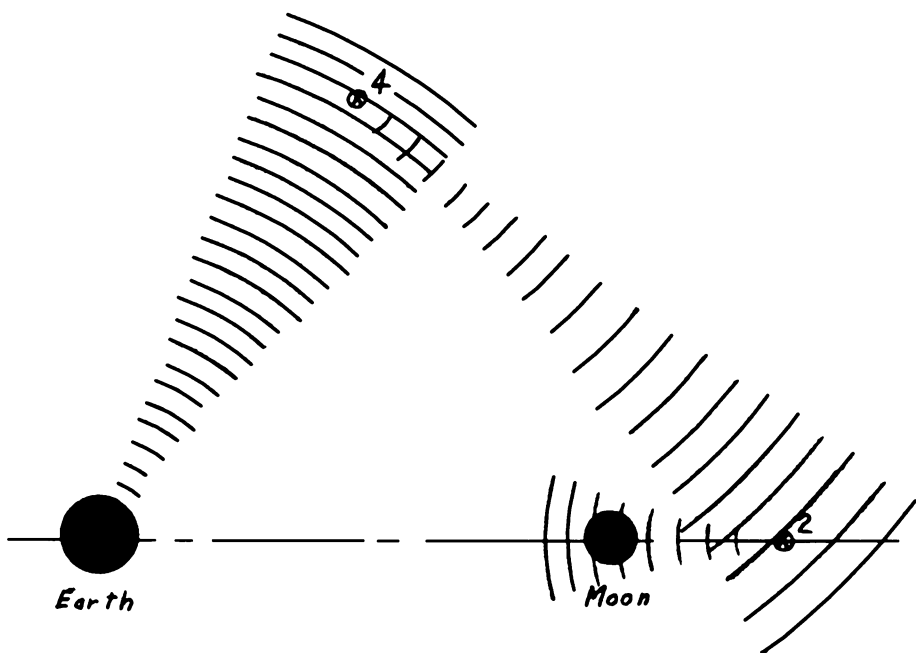


FIG. 2. Communication Network in Earth-Moon System [4].

In 1962, a scheme to use a *single* comsat for lunar farside communications was proposed by G. L. Dunn [5]. This plan called for positioning the comsat about 3200 km off the L2 point perpendicular to the Earth-Moon line. At this location, the comsat would always be in view from the Earth and the Moon's farside. Unfortunately, the stationkeeping cost to maintain a satellite in this position would be approximately 1500 m/sec per year.³

In 1966, a way to keep an L2 satellite in view of Earth-based stations and avoid large stationkeeping costs was proposed by R.W. Farquhar [7]. If certain initial conditions are satisfied, it is possible for a satellite to follow a small periodic orbit around the L2 point. This orbit lies in the Moon's orbital plane and has a period of about 14.47 days. Because the periodic orbit is unstable, stationkeeping is necessary, but the cost can be kept under 10 m/sec per year (i.e., comparable to costs at the L2 point). At the same time, the satellite is allowed to execute a simple-harmonic oscillation normal to the Moon's orbital plane (period ~ 15.30 days). Because the in-plane and out-of-plane oscillation periods are different, the L2 satellite appears to be following a Lissajous trajectory when viewed from the Earth (see Fig. 3). As might be expected, the Lissajous path was not entirely satisfactory because it could not support a continuous communication link. Therefore, to completely eliminate any occultation of the L2 satellite, periodic out-of-plane thrusting was introduced to alter the period of the out-of-plane oscillation so that it would be synchronized with the period of the in-plane oscillation. The cost for the out-of-plane control is roughly

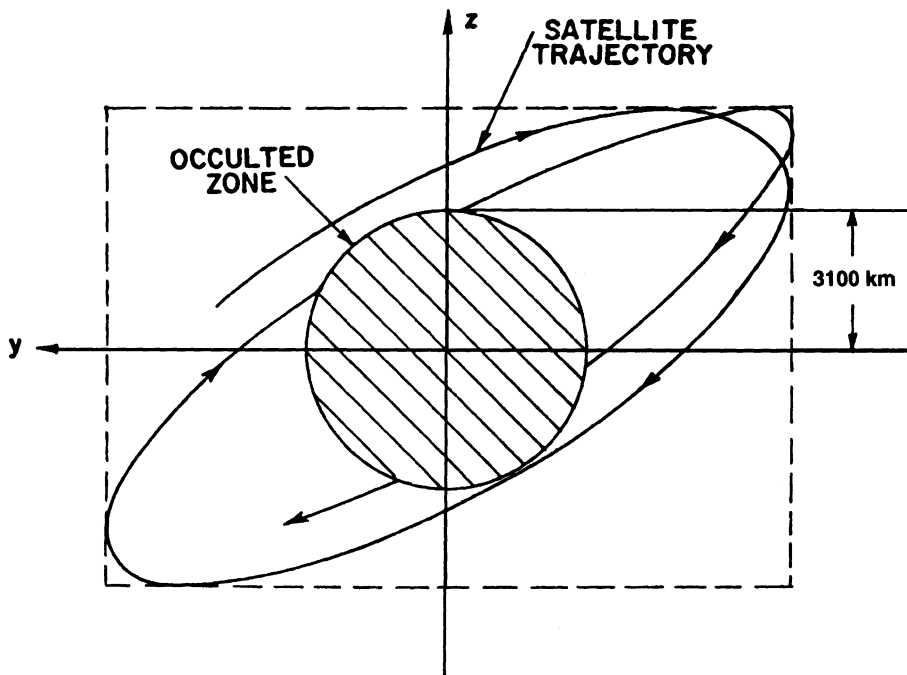


FIG. 3. L2 Satellite Trajectory Viewed from Earth.

³In 1968, the idea of a permanently displaced L2 satellite was re-examined by F. O. Vonbun [6]. In Vonbun's study, it was assumed that the required thrust acceleration would be provided by an electric propulsion system.

100 m/sec per year. The end result was a communication satellite in a “halo orbit” around the translunar libration point that made it possible to maintain a *continuous* communication link between the Earth and the farside of the Moon with a *single* relay satellite (see Fig. 4).

In the years following the first public disclosure of the halo-orbit idea, Farquhar published a number of articles and reports on halo orbits and libration-point satellites [8–16]. The publications covered all aspects of these subjects including station-keeping techniques, trajectories to libration points, orbital control using solar sails and tethers, utilization in post-Apollo lunar operations, and descriptions of several new applications of libration-point orbits. The next important advance in halo orbits came in 1973 when R.W. Farquhar and A. A. Kamel published their discovery of the existence of large halo orbits around the Earth-Moon L_2 point [17]⁴. Farquhar and Kamel found that when the amplitude of the in-plane oscillation is greater than 32,379 km, there is a corresponding value of the out-of-plane amplitude that will produce a path where the fundamental frequencies are equal, producing a “natural” halo orbit. Traces of typical large halo orbits for a one-year period are given in Figs. 5 and 6. Notice that there are two families of this solution (known as Class 1 and Class 2). The in-plane (xy -plane) projection is the same for both families, but the other two projections are reflected on either side of the xy -plane. A more-comprehensive study of halo-orbit families in the Earth-Moon system by J.V. Breakwell and J.V. Brown showed that halo orbits originating in the vicinities of both L_1 and L_2 grow larger, but shorter in period, as they shift towards the Moon [18]. They also found a narrow band of stable orbits roughly half-way to the Moon. Numerical studies by K. C. Howell extended the results of [18] to all values of the mass ratio of the two primaries [19, 20].

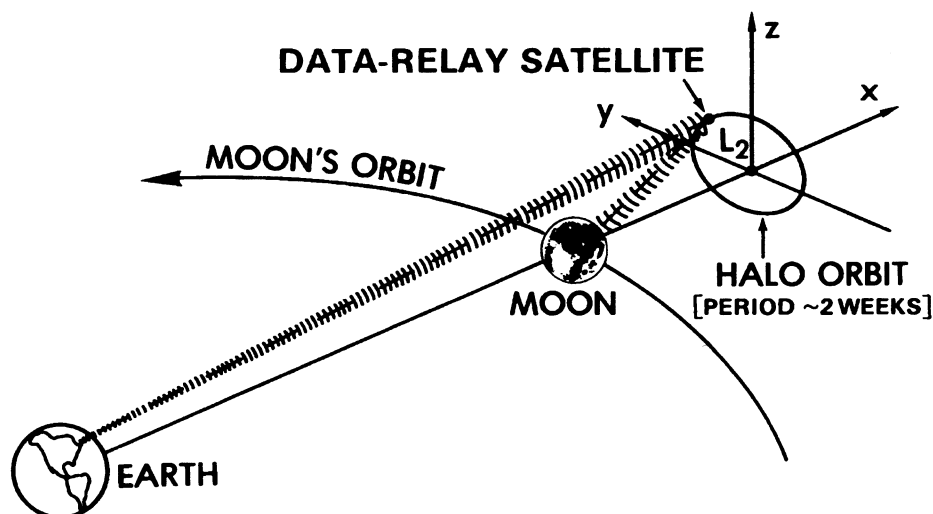


FIG. 4. Lunar Farside Communications Link.

⁴Although this paper concentrated on solutions for the Earth-Moon L_2 point, it was pointed out in the concluding remarks that large halo orbits should exist around both the L_1 and L_2 collinear libration points for any value of the mass ratio of the two primary bodies.

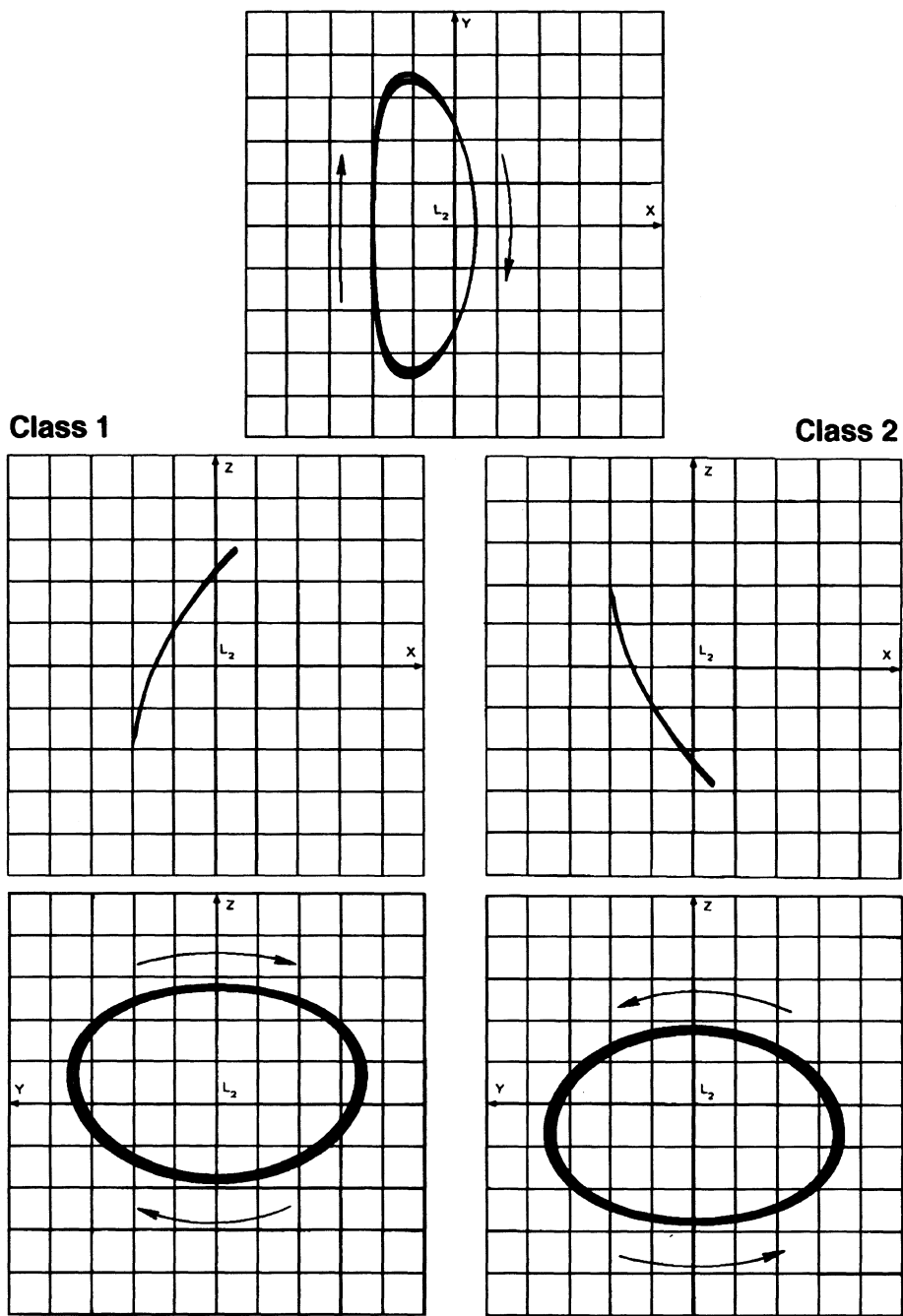


FIG. 5. Large Halo Orbit (Grid Size: 12,900 km) [17].

Finally, as a matter of historical interest, it is worth mentioning that, during the Apollo program, NASA seriously considered placing a communications satellite in a small halo orbit around the translunar libration point. A brief chronology of key events related to this aborted project follows:

- 1964: The desirability of having continuous communications with Apollo vehicles on the far side of the Moon was recognized, but it was decided to ignore the potential risk and hope that catastrophic failures would not occur during communications black-out periods [21].
- January 1969: George M. Low, Apollo Program Manager, receives NASA memo calling his attention to the need for a farside communications link. Mr. Low was not impressed (see Document A-1 in Appendix A).
- January 1970: Goddard Space Flight Center (GSFC) tries to convince an Apollo Steering Committee to submit a request to NASA headquarters for a halo-orbit farside comsat. Apollo group expresses interest, but does not take any action (see Document A-2 in Appendix A).
- April 1970: Apollo-13 emergency causes Apollo Office to re-examine GSFC proposal, but in the end decides to postpone any action.
- Summer 1971: Apollo astronaut Jack Schmitt tries to convince NASA management to select the farside lunar crater Tsiolkovsky as the prime landing site for the Apollo-17 mission that is scheduled for a launch in December 1972.
- September 1971: R. Farquhar of GSFC presents details of the halo-orbit comsat idea to Jack Schmitt and Apollo project personnel. Two contractors, Hughes and TRW, state that they can place a spacecraft in the halo orbit by November 1972. Total cost of project is ~\$60 million (see Document A-3 in Appendix A).
- November 1971: Citing budget concerns, NASA Headquarters rejects the plan for an Apollo-17 landing on the far side of the Moon (see Document A-4 in Appendix A).
- Conclusion: NASA and the U.S. Space Program missed a great opportunity to end the Apollo Program on a high note. It will be a long time before we get another chance to land astronauts on the Moon's hidden face.

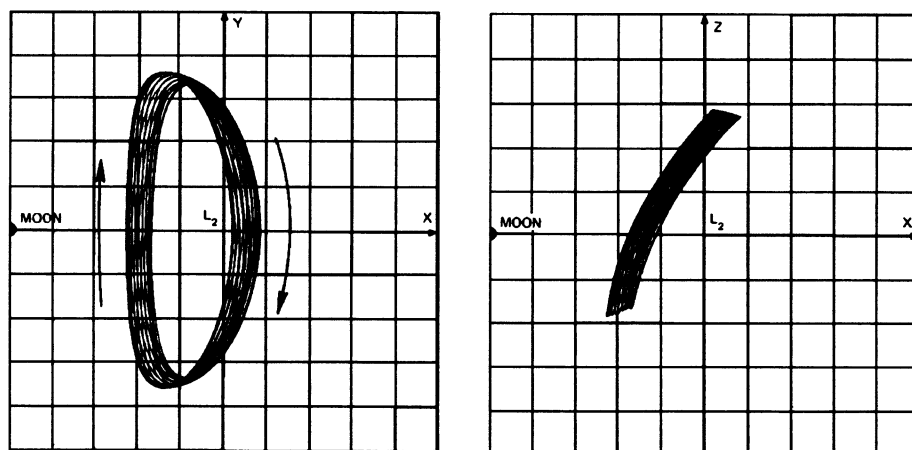


FIG. 6. Large Halo Orbit Relative to Mean L2 Point (Grid Size: 12,900 km) [17].

Solar-Wind Monitor at the Sun-Earth L1 Point

In addition to the five libration points of the Earth-Moon System, there are two Sun-Earth libration points that are located relatively close to the Earth (see Fig. 7). As early as 1964, R. Farquhar recognized that the Sun-Earth L1 point would be an ideal location to continuously monitor the interplanetary environment upstream from the Earth [22]. A suitably instrumented spacecraft placed in the vicinity of this point can provide data on the solar wind about an hour before it reaches the Earth's magnetosphere. However, the spacecraft cannot be stationed too close to the L1 point because the Sun is directly behind this point when viewed from the Earth. This alignment is a problem because the intense solar noise background will severely degrade downlink telemetry. To avoid the zone of solar interference, it is necessary to place the spacecraft in a halo orbit around the L1 point.⁵

Between 1964 and 1970, a number of attempts were made to persuade scientists that a satellite in a halo orbit at the Sun-Earth L1 point would provide valuable and unique data on solar-wind conditions upstream from the Earth. However, there was little or no interest in this proposal until the summer of 1971 when it was presented to N. F. Ness, a renowned space physicist at NASA's Goddard Space Flight Center. The value of a continuous solar-wind monitor near the Sun-Earth L1 point was immediately grasped by Ness, who subsequently championed the effort to send an Explorer-class spacecraft to this location.

In 1972, it was decided to include a Sun-Earth L1 satellite in a proposed three-spacecraft program that would eventually be called the International Sun-Earth Explorer (ISEE) Program [23]. The ISEE Program was a joint project of NASA and the European Space Agency (ESA) that would conduct coordinated investigations in the Earth's magnetosphere and the interplanetary medium. Two of the spacecraft, ISEE-1 and ISEE-2, would reside in a highly elliptical Earth orbit (apogee distance ~ 24 Earth radii). The separation distance between ISEE-1 and ISEE-2 would be relatively small and controlled to allow measurements of the magnetosphere's fine

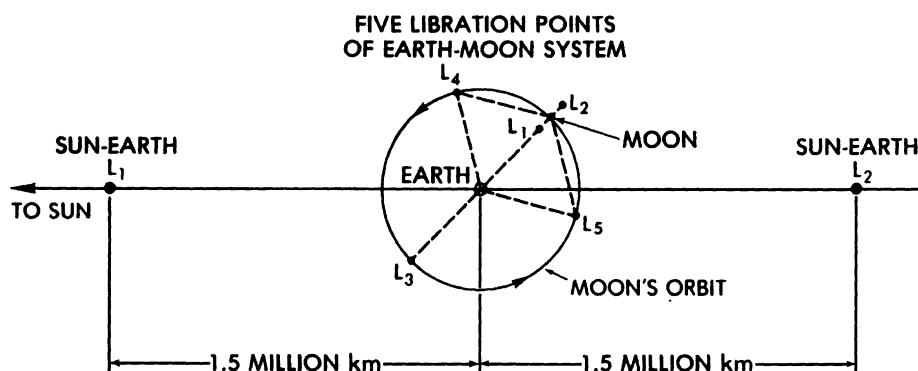


FIG. 7. Libration Points in the Vicinity of the Earth.

⁵An offer to publish a brief description of the L1 solar-wind monitor idea was initiated on October 4, 1966. On this date, a short article by R. Farquhar entitled "Collinear Libration-Point Satellites in the Sun-Earth System" was submitted for possible publication in the AIAA Journal (Log No. J805). Unfortunately, it was summarily rejected with the following comment: "The editors have read the note and state that the subject matter largely has lost its significance as an area of research and yet the subject is being worked to death. The techniques and concepts discussed by the author are by now, well known, and therefore there is no further need to call them to the attention of workers in this field."

structure. The third spacecraft, ISEE-3, would be located in a halo orbit around the Sun-Earth L1 point. From this vantage point, ISEE-3 would be able to monitor interplanetary conditions about one hour before they reached the magnetosphere and ISEE-1 and 2.

ISEE-3 Halo-Orbit Mission

Figure 8 depicts the ISEE-3 spacecraft in its orbital configuration with booms and antennas deployed. The drum-shaped spacecraft is spin stabilized and has a nominal spin rate of approximately 20 rpm. To satisfy a science constraint, the spin axis is oriented perpendicular to the ecliptic plane with an allowable deviation of ± 1 deg. Attitude determination is accomplished with a redundant pair of high-resolution Sun sensors (measurement accuracy ~ 0.1 deg). The spacecraft is equipped with a hydrazine propulsion system for orbit and attitude control. This system has a total of twelve thrusters made up of four radial, four spin-change, two upper-axial, and two lower-axial jets. The fully-redundant thruster configuration permits the spacecraft to thrust in any direction without changing its attitude. Hydrazine fuel is stored in eight conospherical tanks. These tanks contained 89 kg of usable propellant at launch, which corresponds to a ΔV capability of ~ 430 m/sec. The tower structure supports the medium-gain S-band antenna as well as some scientific instrumentation. The medium-gain antenna has a flat, disk-like (“pancake”) pattern that is perpendicular to the spin axis and has an effective beamwidth of 12 deg. Its gain is roughly 7 dB over an isotropic antenna.

A listing of the ISEE-3 science payload is given in Table 1. Detailed descriptions of each of the thirteen experiments can be found in [24]. As stated earlier, the primary science objective of ISEE-3 was to continuously monitor the characteristics of the solar wind and other solar-induced events such as solar flares about an hour before they disturb the space environment near the Earth. Additional objectives included the study of cosmic rays, solar radio bursts, and a wide variety of interplanetary phenomena. One-AU baseline support for experiments on deep-space probes was also provided.

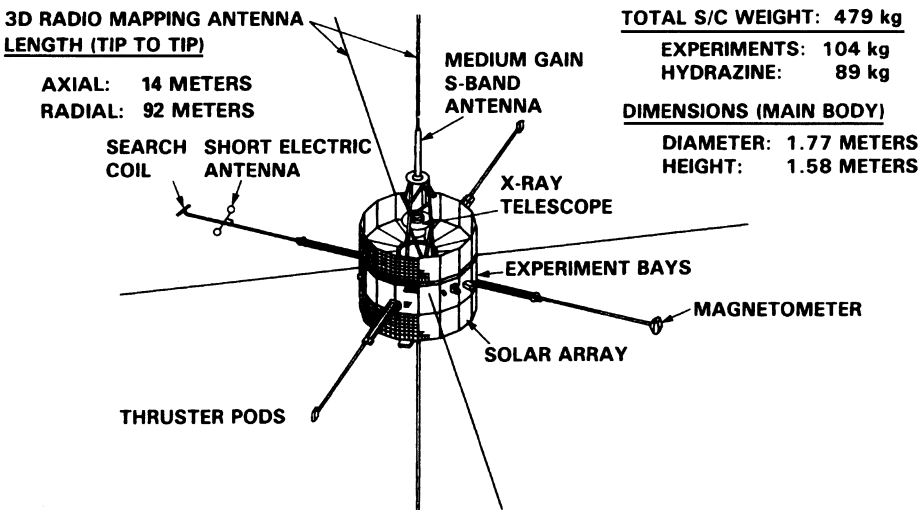


FIG. 8. ISEE-3 Spacecraft.

TABLE 1. ISEE-3 Experiments

Title of Experiment	Principal Investigator	Affiliation
• Solar Wind Plasma	S. J. Bame	Los Alamos National Lab
• Plasma Composition	K.W. Ogilvie	Goddard Space Flight Center
• Magnetometer	E. J. Smith	Jet Propulsion Laboratory
• Plasma Waves	F. L. Scarf	TRW Systems
• Energetic Protons	R. J. Hynds	Imperial College, London
• Radio Waves	J. L. Steinberg	Paris Observatory, Meudon
• X-Rays, Low Energy Electrons	K. A. Anderson	University of California, Berkeley
• Low Energy Cosmic Rays	D. K. Hovestadt	Max Planck Institute
• Medium Energy Cosmic Rays	T. T. Von Rosenvinge	Goddard Space Flight Center
• High Energy Cosmic Rays	E. C. Stone	California Institute of Technology
• High Energy Cosmic Rays	H. H. Heckman	University of California, Berkeley
• Cosmic Ray Electrons	P. Meyer	University of Chicago
• Gamma Ray Bursts	B. J. Teegarden	Goddard Space Flight Center

A rather complete summary of the preflight mission design studies for ISEE-3 is given in [25]. As a result of these studies, a large-amplitude Class-1 halo orbit was selected for ISEE-3's mission orbit around the Sun-Earth L1 point (see Figs. 9 and 10). This orbit passes slightly above and below the ecliptic plane, and easily clears the zone of solar interference ("blackout zone"). Although the RF Sun at S-band subtends an angle of only 0.75 deg., solar noise inputs to the sidelobes of ground antennas will substantially increase the extent of the blackout zone. For a 26-meter antenna at S-band, the blackout zone subtends an angle of about 6 degrees. Notice that the narrow beamwidth of the spacecraft's medium-gain antenna places an upper bound on the out-of-plane amplitude of the halo orbit.

The z-axis amplitude of the halo orbit is 120,000 km, which corresponds to a y-axis amplitude of about 666,670 km. A smaller-amplitude periodic orbit around the

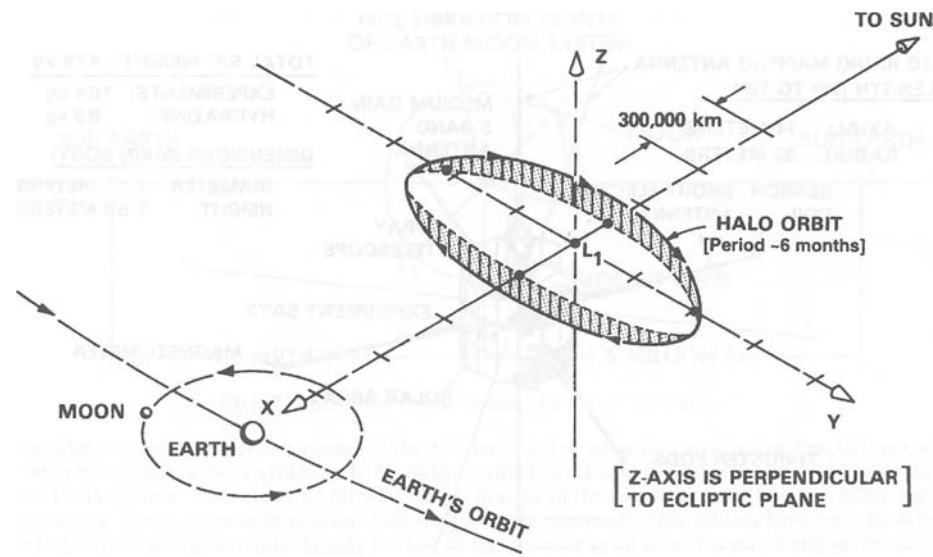


FIG. 9. ISEE-3 Halo Orbit Around the Sun-Earth L1 Libration Point.

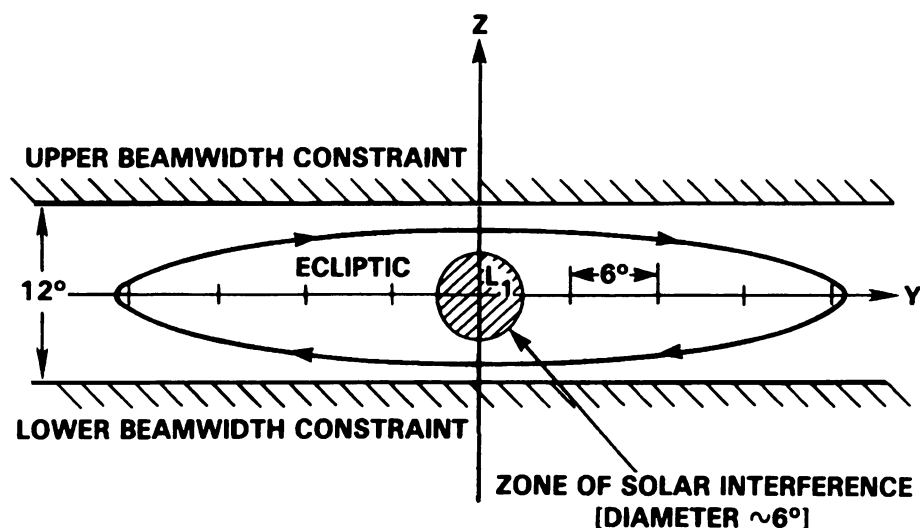


FIG. 10. Halo Orbit as Seen from Earth.

libration point could have been used, but recurring out-of-plane maneuvers (ΔV cost ~ 20 m/sec per year) would have been required to stay out of the blackout zone. More importantly, a large-amplitude halo orbit was preferred because the ΔV requirement for orbit insertion is significantly lower for a large-amplitude orbit. This fact was first reported in [25] where it was shown that the insertion ΔV decreased rapidly as the amplitude of the orbit increased. A graph of the tabulated results in [25] is shown in Fig. 11. It can be seen that the ΔV cost decreases linearly, and goes to zero at an amplitude of approximately 770,000 km.⁶

ISEE-3 was launched towards the halo orbit by Delta rocket #144 on August 12, 1978. Its 100-day transfer trajectory is shown in Fig. 12. Three trajectory maneuvers were used to reach the desired mission orbit. The total ΔV expenditure for these maneuvers was roughly 57 m/sec. It should be noted that the ΔV cost was not fully optimized due to a variety of operational considerations [28].

From November 1978 to June 1982, ISEE-3 remained in its halo orbit, completing four orbits around the L1 point. During the halo-orbit phase, fifteen stationkeeping maneuvers were used for orbital maintenance [29]. The total ΔV cost for these maneuvers was approximately 30 m/sec, which resulted in an average stationkeeping cost of 8.5 m/sec per year.⁷ Attitude and spin control and an anomalous jet firing accounted for an additional 32 m/sec. Even so, in June 1982, ISEE-3 still possessed a ΔV capability of about 310 m/sec.

ISEE-3's science accomplishments during its stay in the vicinity of the L1 point have been reported in a large number of scientific publications. For some examples, see [31–35]. In addition to its purely scientific function, ISEE-3 provided real-time solar wind and interplanetary magnetic field data to the Space Environmental Services Center in Boulder, Colorado beginning in February 1980 [36].

⁶This important result was rediscovered by P. E. Eliasberg in 1986 [26], and again by G. Gomez in 1993 [27].

⁷This number could easily have been reduced to about 3 to 4 m/sec per year under normal circumstances. However, a combination of contradictory orbit determination solutions, and a desire for tight control around the nominal halo orbit led to a higher stationkeeping cost. A description of the stationkeeping technique used by ISEE-3 is given in [30].

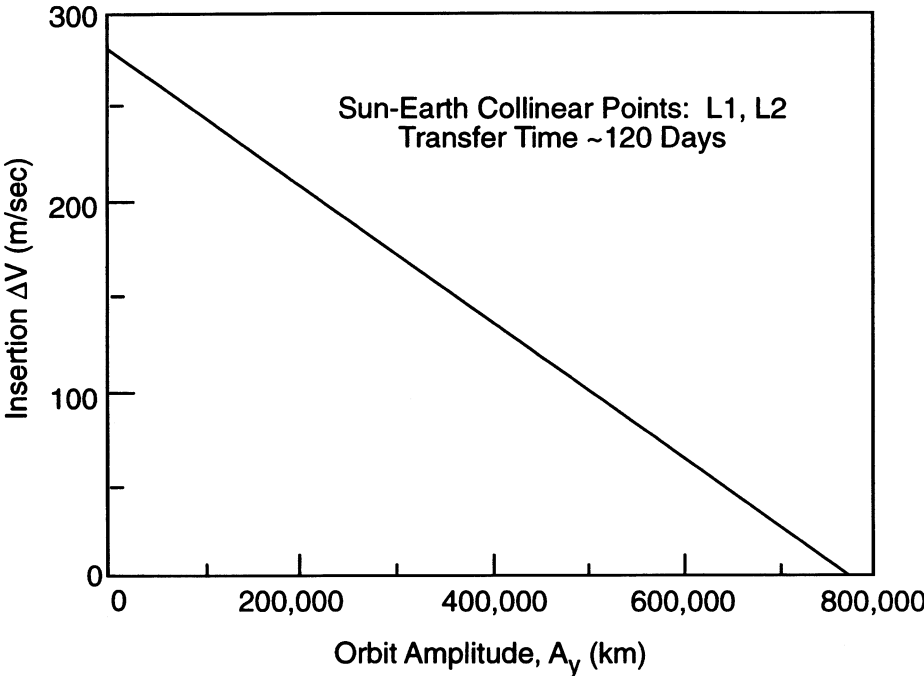


FIG. 11. ΔV Cost for Transfers to Periodic Orbits Around the Sun-Earth Collinear Libration Points ($A_z = 0$).

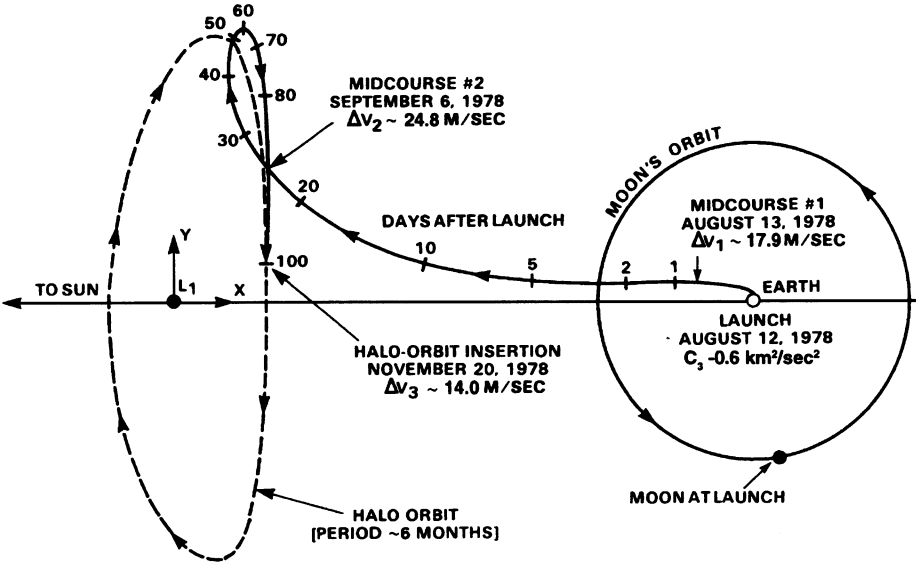


FIG. 12. ISEE-3 Transfer Trajectory to Halo Orbit.

Proposal to Send ISEE-3 to the Distant Geotail Region and on to a Comet

Even before the launch of ISEE-3, various options for an extended mission were discussed. One of the leading candidates for an extended mission was to move ISEE-3 from the L1 point to the corresponding L2 point deep in the Earth's magnetic tail. A sketch of this trajectory was prepared in 1974 and circulated to the ISEE investigators in 1975 (see Fig. 13). The possibility of sending ISEE-3 to a nearby comet was also considered. However, the comet option was quickly discarded because it was erroneously concluded that useful data from ISEE-3 could not be returned from Earth distances greater than 0.04 AU.⁸

In early 1981, however, there was little interest in any kind of an extended mission for ISEE-3. An overwhelming majority of the ISEE Science Team wanted ISEE-3 to remain in its halo orbit for the foreseeable future. However, Fred Scarf, the principal investigator for ISEE-3's plasma wave experiment, had other ideas. Fred was becoming somewhat bored with ISEE-3's monitoring function, and wanted to use the spacecraft for more exciting science objectives such as the exploration of the distant geomagnetic tail, and perhaps a flyby of Halley's comet. Therefore, in March 1981, Fred Scarf telephoned R. Farquhar to see if something could be done to change ISEE-3's mission. Fred reiterated his interest in a geotail mission, and then asked the following question: "Bob, you have said that it would be possible to retarget ISEE-3 for an encounter with Halley's comet in 1986. Is it possible to send ISEE-3 through Halley's comet and then return it to an Earth orbit? If this were possible, we

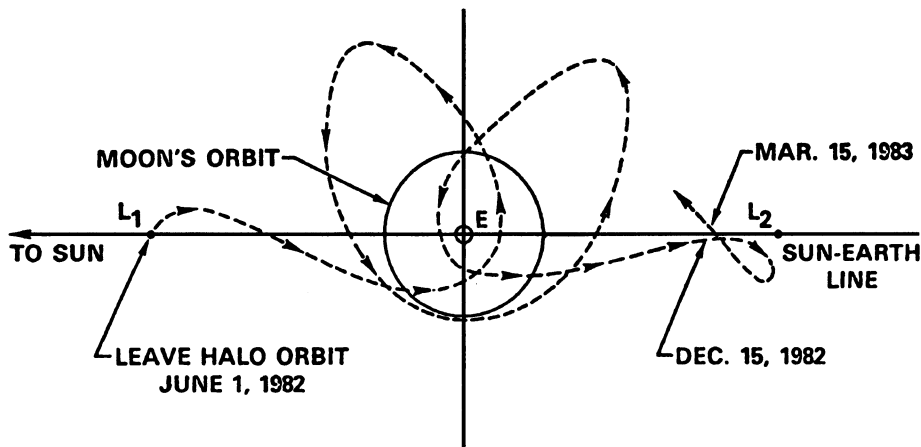


FIG. 13. 1974 Drawing of ISEE-3 Extended Mission to Distant Geomagnetic Tail.

⁸ISEE-3 did not have a data-storage system, and could only provide data while in contact with ground stations. It routinely transmitted 2000 bits/sec into a 26-meter "standard" S-band antenna at a distance of 0.01 AU. Therefore, it was assumed that a maximum of 500 bits/sec could be sent from 0.04 AU if a 64-meter ground antenna was used. However, this assumption was way off the mark because it underestimated the greatly-increased performance of the low-noise Deep-Space Network (DSN) antennas. This crucial fact was first pointed out by Joel Smith and Warren Martin at the Jet Propulsion Laboratory (JPL) in July 1981. Using this fact and other creative ideas (arraying ground stations, operating two spacecraft transmitters simultaneously, improved decoder, etc.), Warren Martin was able to show that ISEE-3 had enough link margin to support a data rate of 1000 bits/sec at a distance of 0.5 AU [37].

would not only be able to obtain good data during the Halley encounter, but we might also be able to return the spacecraft to Earth orbit covered with Halley dust.”

Although Fred’s original proposal was rather ambitious, it eventually led to a flight plan that was adopted by the ISEE-3 team. A chronology of some of the key events in the development and approval of ISEE-3’s extended mission follows:

- March 1981: R. Farquhar and F. Scarf devise a plan that would send ISEE-3 into the geotail region and then use lunar gravity-assist maneuvers to reach comet Halley. Data from ISEE-3’s encounter with Halley would be relayed to Earth through a Japanese spacecraft that would arrive at Halley at the same time. ISEE-3 would use an eight-year Earth-return trajectory ($C_3 \sim 1.83 \text{ km}^2/\text{sec}^2$) that leaves Earth on October 17, 1984 and arrives at Halley on March 11, 1986.⁹
- April 27, 1981: NASA Headquarters authorizes a study to determine the feasibility of retargeting the ISEE-3 spacecraft for an encounter with Halley’s comet (see Document B-1 in Appendix B).
- July 1981: Joel Smith and Warren Martin of JPL show that it is possible to obtain useful science data from ISEE-3 at distances as far as one AU from Earth.
- August 1981: A study to investigate the feasibility of moving ISEE-3 to the geomagnetic tail is initiated (see Document B-2 in Appendix B).
- September 1981: Plans for a dedicated U.S. mission to Halley’s comet are abandoned. Fred Scarf asks R. Farquhar to re-examine the possibility of sending ISEE-3 to a comet.

The ISEE-3 science payload was not ideal for cometary investigations. It did not have any imaging instruments or dust detectors and was heavily oriented towards cosmic ray studies. Nevertheless, at least six ISEE-3 experiments (solar-wind plasma, plasma composition, magnetometer, plasma waves, energetic protons, and radio waves) would provide useful data on the interaction of the solar wind with a comet, a chief objective of cometary exploration.

The existence of an acceptable comet mission opportunity for ISEE-3 was not obvious. A number of rather stringent criteria had to be satisfied before a candidate mission option could be certified. For instance, the orbital energy required to escape from the Earth-Moon system and reach the comet would have to come almost entirely from a lunar gravity-assist maneuver. Propulsive maneuvers could also be used, but large perigee burns were out of the question.¹⁰ Because the post-swingby energy derived from a lunar flyby is somewhat limited, comet mission opportunities with C_3 requirements greater than $3 \text{ km}^2/\text{sec}^2$ were not considered. Other constraints specified that the perihelion distance of the transfer trajectory should be greater than 0.90 AU and the aphelion distance should not exceed 1.20 AU. These restrictions were dictated by the thermal and power characteristics of the ISEE-3

⁹The plan to send ISEE-3 into the distant geotail was realistic and workable. However, the bizarre scheme to use a Japanese relay spacecraft to transmit ISEE-3’s Halley data back to Earth was never a serious proposal. It was merely a ploy to get NASA management interested in a dedicated Halley sample-return mission [38]. In March 1981, we believed that it was not possible to communicate directly with ISEE-3 at distances greater than 0.04 AU.

¹⁰Perigee maneuvers for ISEE-3 would be performed with the radial thrusters in a pulsed mode. Because the duty cycle was only 12.5%, the finite-burn losses would have been unacceptably high for long maneuvers.

spacecraft. To ensure adequate communications margins, the Earth distance at the comet encounter had to be under one AU.

Mission opportunities with the aforementioned limitations are quite rare, but the 1985-86 period was an exception. Two viable options were identified, one to comet Giacobini-Zinner in September 1985, and the other to Halley’s comet in March 1986. The key parameters for these mission opportunities are given in Table 2. Note that the launch dates for both options are only a month apart.

The initial reaction was to go for Halley’s comet. However, it soon became apparent that the Giacobini-Zinner option had several advantages, the most important being a shorter communications distance at encounter.¹¹ Of course, political factors also played a role in selecting the Giacobini-Zinner option. The September 1985 encounter at Giacobini-Zinner would occur about six months before spacecraft from the Soviet Union, the European Space Agency, and Japan were scheduled to intercept Halley. By sending ISEE-3 to Giacobini-Zinner, the United States would be the first nation to obtain *in-situ* measurements at a comet.

Once it was decided that the target would be Giacobini-Zinner, it was necessary to obtain details of the ISEE-3 trajectory and the required lunar gravity-assist maneuver. The utilization of a lunar gravity-assist maneuver to escape from the Earth-Moon system and enter a heliocentric orbit had been done before. As early as 1959, a lunar flyby catapulted the Soviet space probe Luna-1 into a heliocentric orbit. However, using the Moon to target a spacecraft for an encounter with a comet requires greater skill. The added complexity of a targeted lunar swingby is fairly

TABLE 2. Comet Mission Options*

	Halley	Giacobini-Zinner
Launch Opportunity	January 1984	December 1983
Maximum Launch Energy for 30-Day Window (km ² /sec ²)	2.63	2.67
Typical Mission Parameters		
Launch Date	January 15, 1984	December 15, 1983
Launch Energy-C ₃ (km ² /sec ²)	2.42	2.48
Comet Encounter Date	March 16, 1986	September 11, 1985
Sun Distance (AU)	0.93	1.03
Earth Distance (AU)	0.91	0.47
Phase Angle (Degrees)	112.4	96.6
Flyby Speed (km/sec)	70.1	20.7
Spacecraft Trajectory		
Perihelion (AU)	0.92	0.93
Aphelion (AU)	0.98	1.03
Inclination (Degrees)	2.7	0.1

*Data based on two-body patched-conic calculations

¹¹The Giacobini-Zinner option also had another totally fortuitous communications advantage. It was noticed that, on September 11, Giacobini-Zinner would pass almost directly overhead as viewed from the 305-meter antenna located at Arecibo, Puerto Rico. Sometime later, an S-band receiver was added to the Arecibo antenna, and this antenna was used to augment DSN coverage of the Giacobini-Zinner encounter.

obvious, because in this situation, the aim point in the impact plane, the relative velocity vector between the spacecraft and the Moon, and the timing of the lunar flyby, are precisely related.

To determine the required lunar-swingby conditions, the trajectory was integrated backwards in time, starting at the comet encounter. It did not take long to find that a single lunar gravity-assist maneuver would be able to transform the heliocentric trajectory into a high-energy Earth orbit that reached the geotail in 1983 (see Fig. 14). However, the altitude of the lunar swingby at S is only 67 km. This rather unsettling result showed that the Moon's gravity was barely able to provide the energy boost needed to reach Giacobini-Zinner.¹²

Returning now to the chronology of key events:

- December 21, 1981: NASA Headquarters authorizes feasibility studies of extended-mission alternatives for ISEE-3 including an option to intercept comet Giacobini-Zinner (see Document B-3 in Appendix B).
- February 8, 1982: The plan to redirect ISEE-3 to Giacobini-Zinner is presented to the ISEE Science Working Team. A majority of the ISEE-3 investigators (7 out of 13) vote to return ISEE-3 to the L1 halo orbit after its geotail excursion.
- May 14, 1982: The Giacobini-Zinner plan is supported by the Comet and Asteroid Working Group of the Solar System Exploration Committee (see Document B-4 in Appendix B).
- June 10, 1982: A ΔV maneuver of only 4.5 m/sec is used to initiate ISEE-3's transfer to the geotail (see Fig. 15). To retain the option to redirect ISEE-3 to Giacobini-Zinner, the orbital period of the figure-eight loop was deliberately chosen to take ISEE-3 to the vicinity of the Moon at S1 (see Document B-5 in Appendix B).

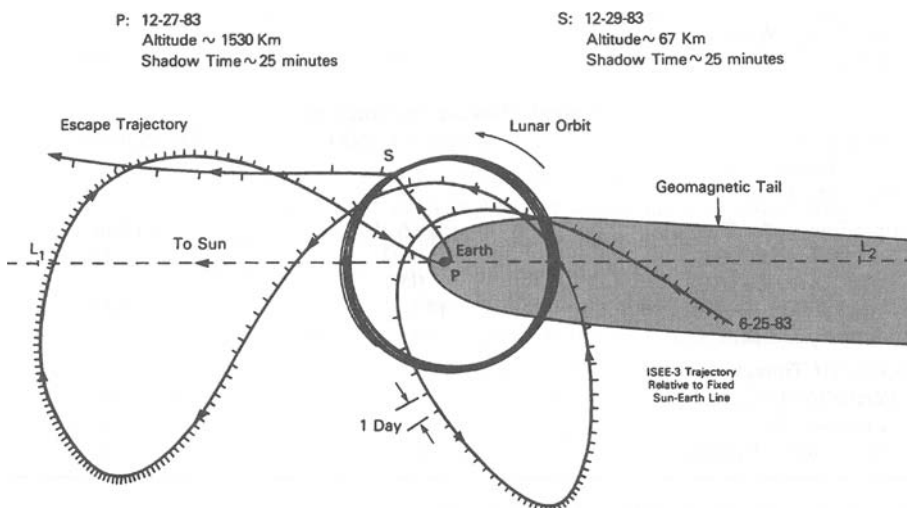


FIG. 14. ISEE-3 Escape Trajectory (Mode 1).

¹²This result was rather surprising because an earlier calculation using a two-body patched-conic model yielded a swingby altitude of about 840 km.

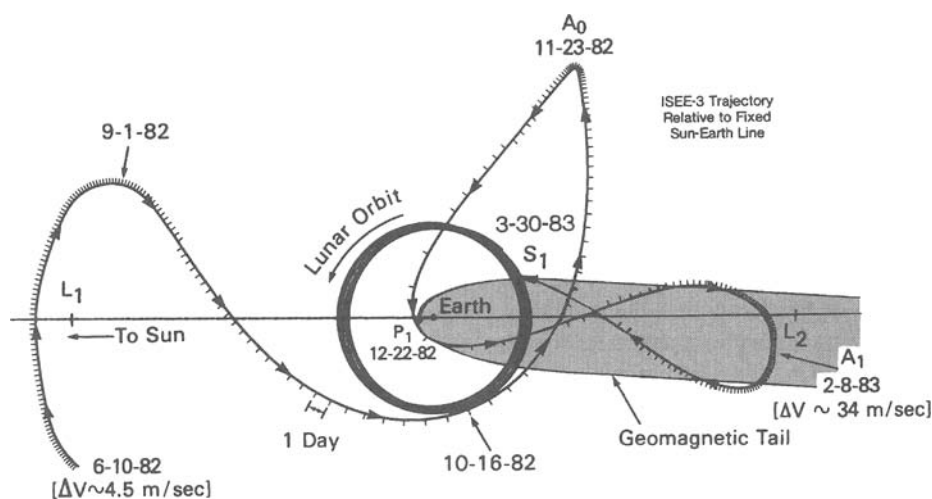


FIG. 15. Transfer from Halo Orbit to Geomagnetic Tail.

- June 15, 1982: The plan to send ISEE-3 into the geotail and then on to comet Giacobini-Zinner is endorsed by the Space Science Board of the National Academy of Sciences [39].
- June 29, 1982: ISEE-3's geotail mission is approved by NASA Headquarters.
- August 6, 1982: ISEE-3 extended-mission options are presented to the Office of Space Science and Applications at NASA Headquarters.
- August 30, 1982: The mission of ISEE-3 to explore Giacobini-Zinner is approved by NASA Headquarters [40] (see Document B-6 in Appendix B).

The struggle to gain approval for the ISEE-3 comet mission was not easy. There was strong opposition from several ISEE-3 investigators including the Project Scientist, Tycho von Rosenvinge. However, the endorsement of the comet proposal by the Space Science Board was decisive, and NASA management dutifully followed their recommendations. Finally, it should be mentioned that two ISEE-3 investigators, Fred Scarf and Ed Smith, were very effective in presenting scientific arguments in favor of the comet option. Their advocacy was crucial in obtaining a favorable recommendation from the Space Science Board.

Constructing a Solution with Double Lunar-Swingby Trajectory Segments

At the same time that NASA's approval of the extended mission to Giacobini-Zinner was solicited, the technical problem of linking the trajectories of Figs. 14 and 15 was examined. Initially, it was assumed that a satisfactory solution would be found with relative ease. However, the extremely complex nature of the problem soon became apparent. Although a large variety of propulsive and lunar gravity-assist maneuvers could be used to link the swingbys on March 30, 1983 (S_1 in Fig. 15) and December 29, 1983 (S in Fig. 14), the determination of an achievable maneuver sequence was not obvious. The difficulty of the problem was compounded by the fact that two-body patched-conic analyses were almost useless. Because the ISEE-3

flight path is strongly perturbed by the Sun's gravitational field, it was necessary to test fully-integrated trajectories for acceptable solutions. Unfortunately, this trial-and-error procedure was rather time consuming, sometimes taking as long as two weeks to check out a particular maneuver sequence. In early 1982, there was still no clue in finding a way to connect the swingbys of Figs. 14 and 15. Therefore, it was decided to abandon the escape trajectory of Fig. 14 and try an entirely new approach.

As shown in Fig. 16, the new approach uses a fundamentally different type of escape trajectory. In Fig. 16, the altitude of the final swingby is higher than in Fig. 14 because this swingby is located very close to the Moon's perigee. Another important feature of Fig. 16 is that the period of the twisting orbital segment was chosen so that the trajectory re-encounters the Moon at S_0 . The swingby at S_0 was included to facilitate a more systematic procedure in finding a way to link the trajectory of Fig. 15 with an escape trajectory to comet Giacobini-Zinner. The key ingredient of this systematic process was the use of the recently-discovered Double Lunar-Swingby (DLS) trajectory concept [41].

The top section of Fig. 17 illustrates the basic DLS trajectory technique. Assume that a spacecraft is initially located near the apogee of the smaller orbit at A_1 . After its next perigee passage, the natural orbital precession will position the spacecraft for a trailing-edge swingby of the Moon at S_1 . The swingby maneuver at S_1 will then rotate the line of apsides back to the Sun-Earth line and will also raise the apogee distance to A_2 . A leading-edge swingby at S_2 after the Moon has completed one full orbit plus the S_1S_2 segment will return the spacecraft to its original orbit. This sequence of orbit pairs could be repeated indefinitely or, by slightly changing the swingby conditions at S_1 and S_2 , the spacecraft could be placed into different periodic orbits as shown in the other sections of Fig. 17. The three classes of orbits shown in Fig. 17 represent just a few of the many solutions that can be formed with the DLS technique. Additional solutions can be obtained by increasing the time interval as well as the number of orbital loops in the inner trajectory segment (e.g.,

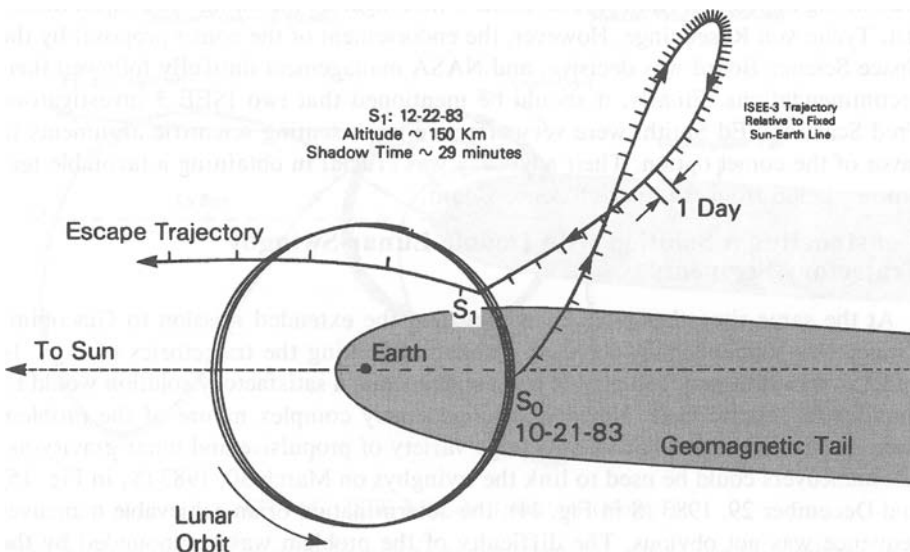


FIG. 16. ISEE-3 Escape Trajectory (Mode 2).

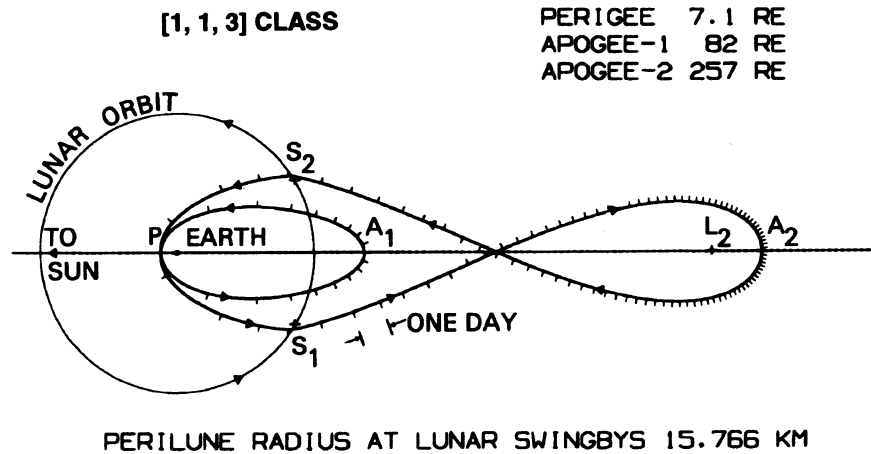
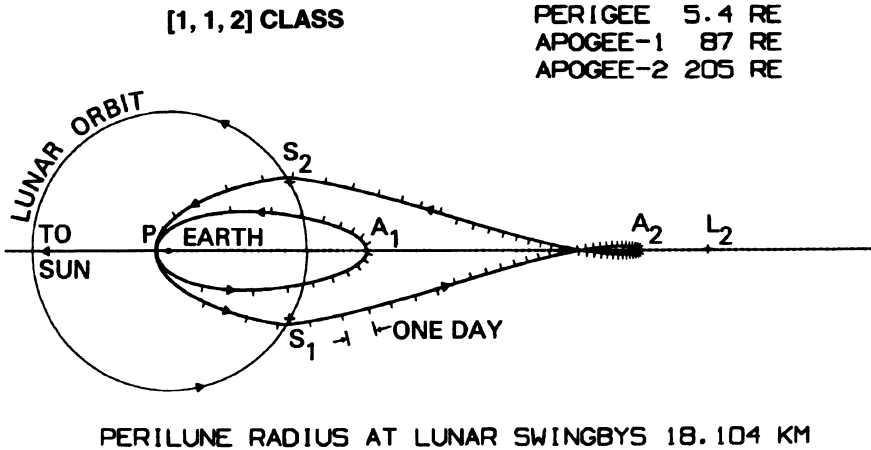
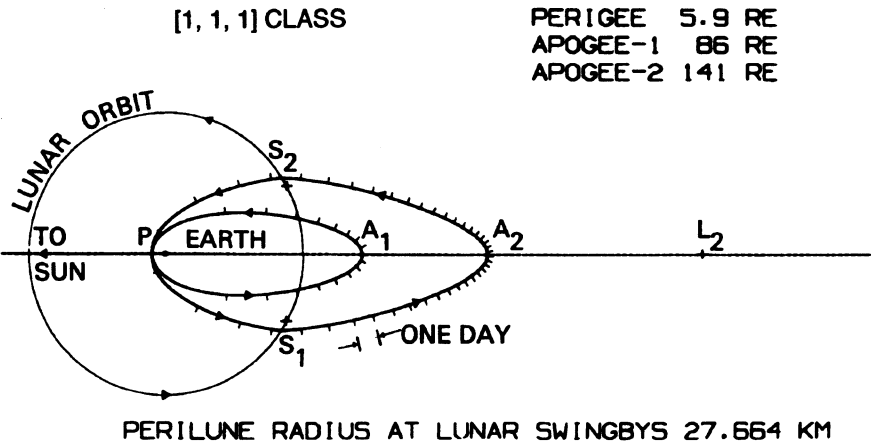
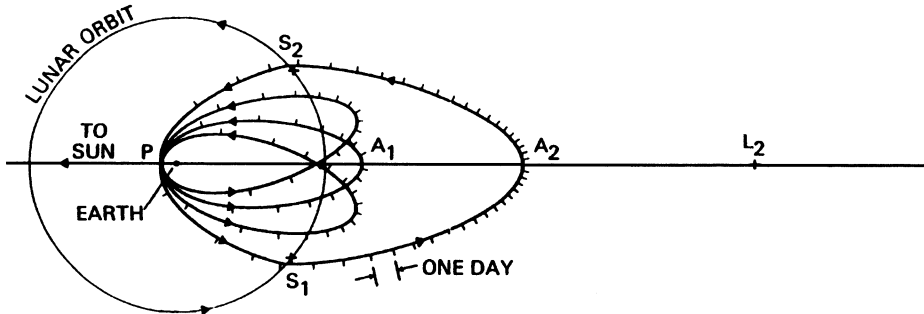


FIG. 17. Sun-Synchronous Periodic Orbits Using the Double Lunar-Swingby (DLS) Technique.

see Figs. 18 and 19). Four, five, six and seven-month outer loops are also possible. DLS trajectories are classified by using the shorthand notation, $[A, B, C]$, where A is the approximate number of months between lunar swingbys for the inner trajectory segment, B is the number of complete loops for the inner trajectory segment, and C is the approximate number of months between swingbys for the outer trajectory segment.

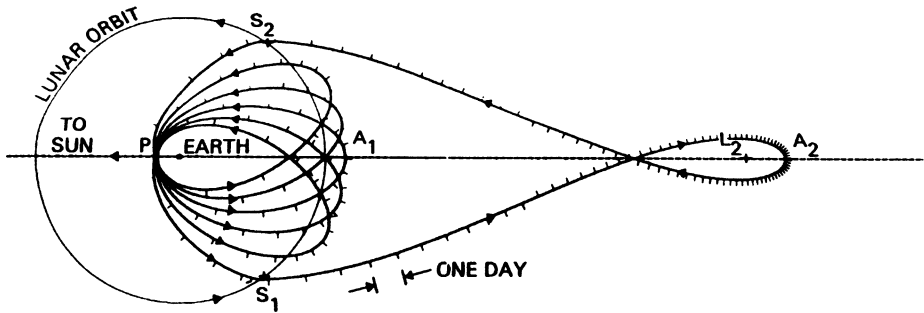
The periodic orbits in Figs. 17–19 have been generated with a two-body patched-conic dynamical model. When a more-realistic model that includes the effects of solar perturbations and the Moon's orbital eccentricity is used, the symmetrical shapes are distorted and the perigee, apogee, and swingby distances are different. Nevertheless, the average apsidal rotation can be controlled to the required Sun-synchronous rate.

Return now to the problem of finding a trajectory that would allow ISEE-3 to utilize the escape trajectory in Fig. 16. To simplify the search, it was decided to fix the initial three-month loop in Fig. 15 and the final two-month loop in Fig. 16. With this constraint, the problem was reduced to finding an acceptable DLS sequence between S_1 in Fig. 15 and S_0 in Fig. 16. Thus, it was relatively easy to identify candidate DLS sequences for the seven-month interval between the two fixed lunar swingbys. Two of the many worksheets that were used to define possible DLS sequences are illustrated in Fig. 20. The integers in the figure represent the number



PERILUNE RADIUS AT LUNAR SWINGBYS 19.936 KM

FIG. 18. $[2,3,1]$ DLS Trajectory.



PERILUNE RADIUS AT LUNAR SWINGBYS 12.891 KM

FIG. 19. $[3,5,3]$ DLS Trajectory.

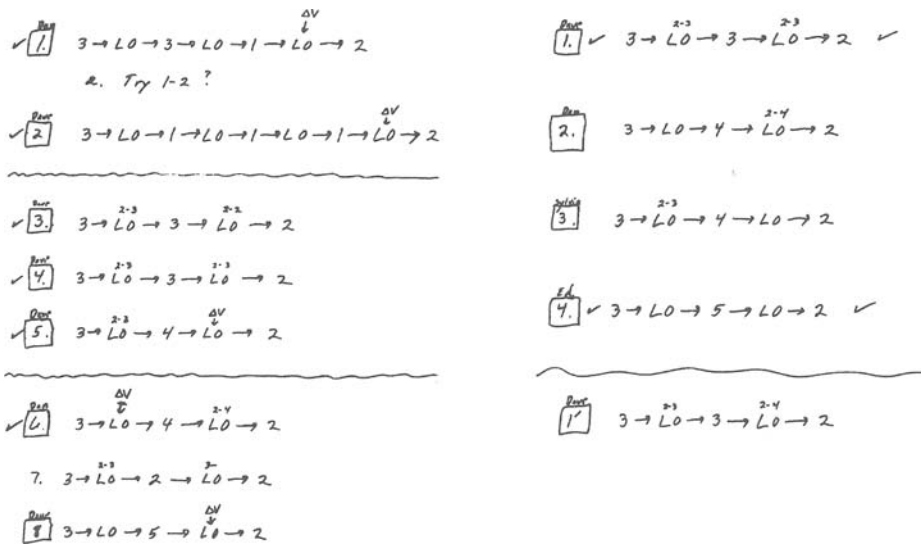


FIG. 20. Worksheets Used to Define Possible DLS Sequences.

of months between lunar encounters for the outer-loop portions of the DLS sequences. The symbol LO signifies an inner-loop segment with one month between encounters. A superscript (m-n) over the LO symbol is used to denote an inner-loop segment requiring m months between lunar swingbys with n complete orbital revolutions during that period. For example, the $3 \rightarrow LO \rightarrow 4 \rightarrow LO(2-4) \rightarrow 2$ sequence begins with a three-month outer loop, followed by a one-month, one-revolution inner loop, followed by a four-month outer loop, followed by a two-month, four-revolution inner loop, and ending with a two-month outer loop. The total number of months in each sequence is twelve, including the initial three-month geotail traverse and the final two-month escape trajectory of Fig. 16.

After considerable anguish and tribulation (see Appendix C), a satisfactory solution was found [42]. This solution is the $3 \rightarrow LO \rightarrow 5 \rightarrow LO \rightarrow 2$ sequence and is shown in Figs. 15, 21, and 22. A total of five lunar swingbys and four planned

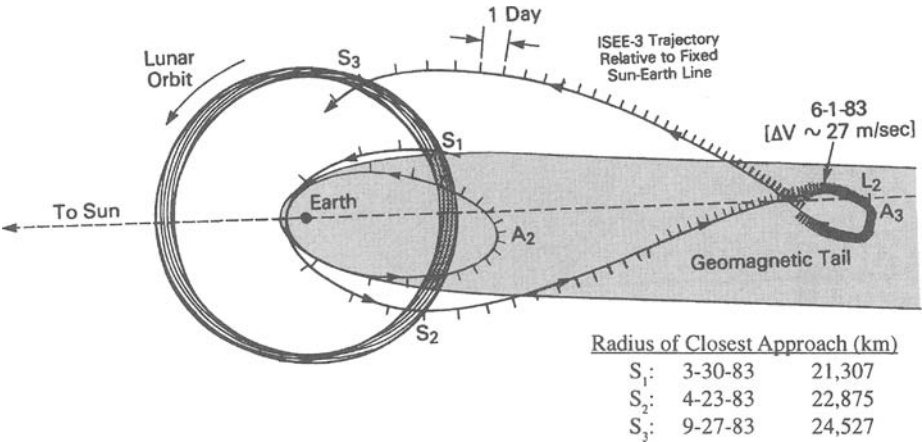


FIG. 21. Five-Month Geotail Excursion.

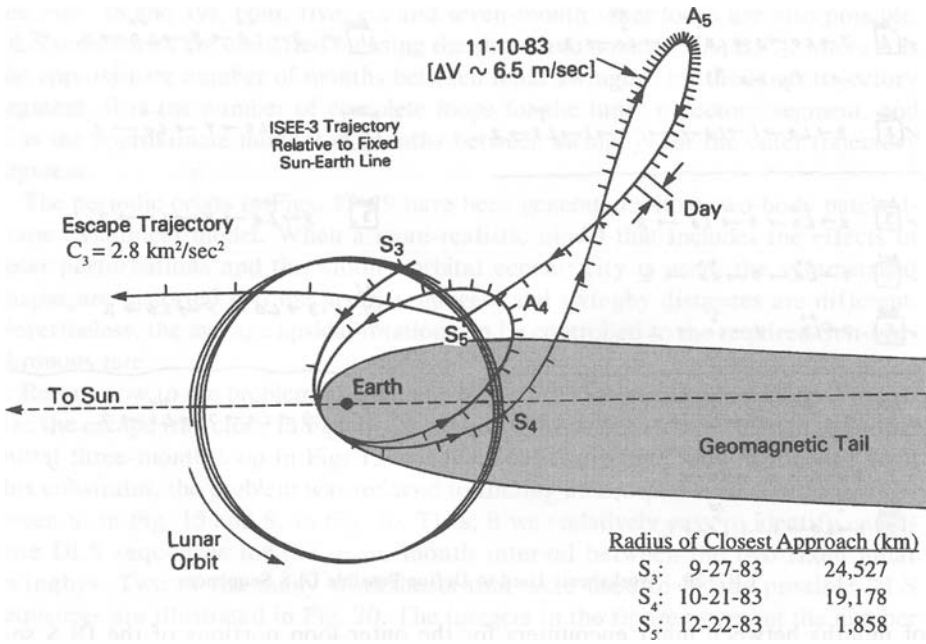


FIG. 22. ISEE-3 Escape Trajectory.

propulsive maneuvers were used to accomplish the selected sequence. It should be mentioned that the final design was further complicated by the need to avoid shadows of the Earth and the Moon. Additional details of the ISEE-3 trajectory can be found in [42–44].

Fifteen propulsive maneuvers were used to carry out the transfer from the halo orbit to the escape trajectory (see Table 3). The four planned maneuvers accounted

TABLE 3. ISEE-3 Propulsive Maneuvers (1982–83)

Maneuver Date	ΔV (m/sec)		
	Radial	Axial	Total
*6-10-82	4.50	—	4.50
8-10-82	0.45	—	0.45
12-15-82	0.12	—	0.12
*2-8-83	—	34.07	34.07
3-1-83	0.33	−0.79	1.12
3-22-83	0.17	0.19	0.36
4-2-83	0.07	—	0.07
*6-1-83	17.87	9.16	27.03
7-1-83	1.52	0.69	2.21
8-12-83	0.20	0.11	0.31
9-20-83	0.03	—	0.03
10-1-83	0.16	—	0.16
10-12-83	—	0.15	0.15
*11-10-83	0.38	6.11	6.49
11-23-83	0.10	0.06	0.16
			77.23

*Planned Maneuvers

for 72 m/sec of the total ΔV cost of 77 m/sec. Maneuver execution errors and orbit-determination uncertainties were responsible for the remaining 5 m/sec. Optimization of the planned maneuvers is discussed in [44]. Details of navigational corrections are covered in [45].

The final lunar swingby on December 22, 1983 is shown in Figs. 23 and 24. During this swingby, ISEE-3 was in the Moon's shadow for twenty-eight minutes, and could not communicate with ground stations for almost one hour. Because ISEE-3's battery had failed in 1981, there was some concern that the hydrazine fuel would freeze during the lunar shadow. Fortunately, the temperature drop on the spacecraft was smaller than predicted, and ISEE-3 came through the lunar shadow in good shape [46].

Before leaving this section, it should be noted that the lengthy distant-tail trajectories shown in Figs. 15 and 21 allowed ISEE-3 to obtain a voluminous amount of scientific data from this previously unexplored region. Reviews of ISEE-3's observations in the distant geotail are given in [47–51].

Encounter with Comet Giacobini-Zinner

Immediately following the lunar gravity-assist maneuver on December 22, 1983, NASA announced that it had decided to change the name of ISEE-3 to the "International Cometary Explorer (ICE)." Among other things, this name change denoted the spacecraft's transformation from an Earth orbiter to an interplanetary probe.

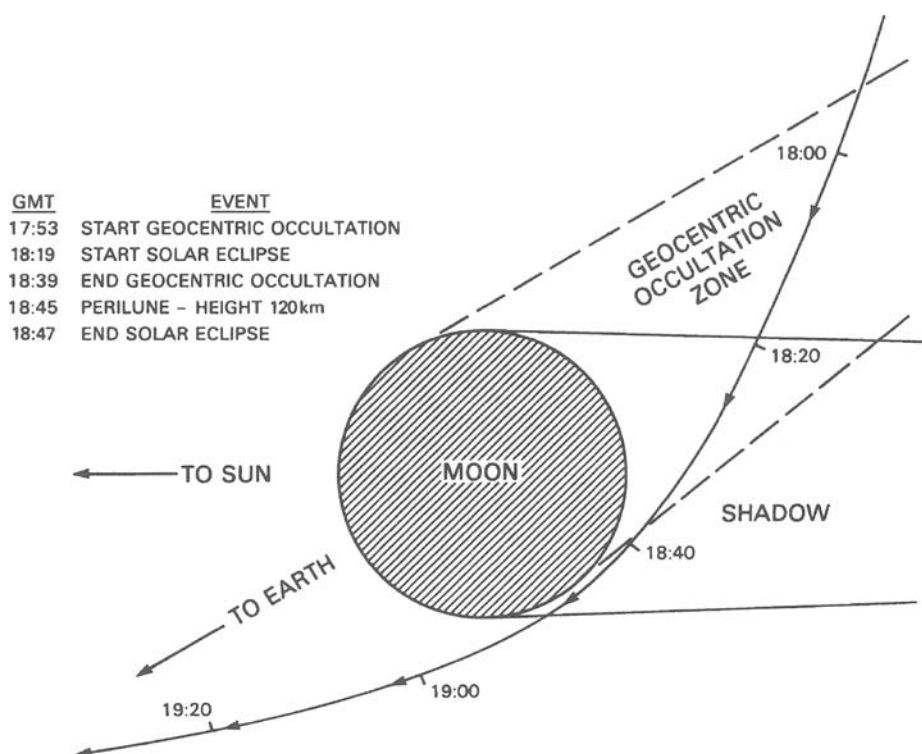


FIG. 23. ISEE-3 Lunar Swingby, December 22, 1983.

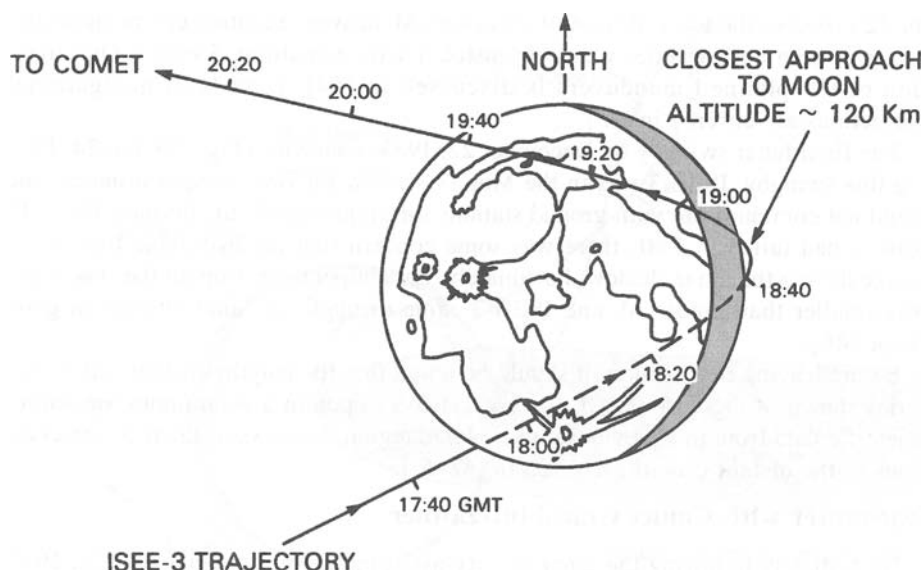


FIG. 24. Trajectory of ISEE-3 Around the Moon as Seen from the Earth.

Figures 25 and 26 illustrate the heliocentric trajectory of the newly-christened Cometary Explorer. The looping motion depicted in Fig. 26 is a direct result of plotting the trajectory with respect to a fixed Sun-Earth line. In Fig. 25, it can be seen that the encounter with Giacobini-Zinner on September 11, 1985 takes place as the comet crosses the ecliptic plane. This, by the way, is only six days after the comet's perihelion passage.

Although the final lunar gravity-assist maneuver had placed ICE into a near-nominal heliocentric trajectory, there was some concern that Giacobini-Zinner would not arrive at the planned intercept point on schedule. Giacobini-Zinner was poorly observed at its previous apparition in 1979, and it was conceivable that its arrival time would be in error by as much as one day. An error of this magnitude, if it was not discovered before 1985, would have required a large ΔV maneuver to retarget ICE to the new arrival time. As a matter of fact, the ΔV cost to correct a one-day error in April 1985 would have exceeded ICE's remaining ΔV capability of roughly 233 m/sec. Therefore, there was considerable relief when Giacobini-Zinner was observed at the Kitt Peak National Observatory on April 3, 1984, and it was verified that the comet would reach the intercept point within 0.07 days of the predicted time [52].

Because the comet's orbital deviations were manageable, it was decided to postpone any major retargeting maneuver until sometime in 1985 when improved comet orbit solutions would be available. It was also necessary to finalize the aim point in the comet's tail. Because the motion of the "wagging tail" in response to solar-wind velocity variations becomes more pronounced farther from the nucleus, an aim point far down the tail carried the increased risk of missing it altogether. On the other hand, aiming too close to the nucleus might create a potentially serious dust hazard. As a compromise, the intercept point in the tail was chosen to be 10,000 km from the nucleus.

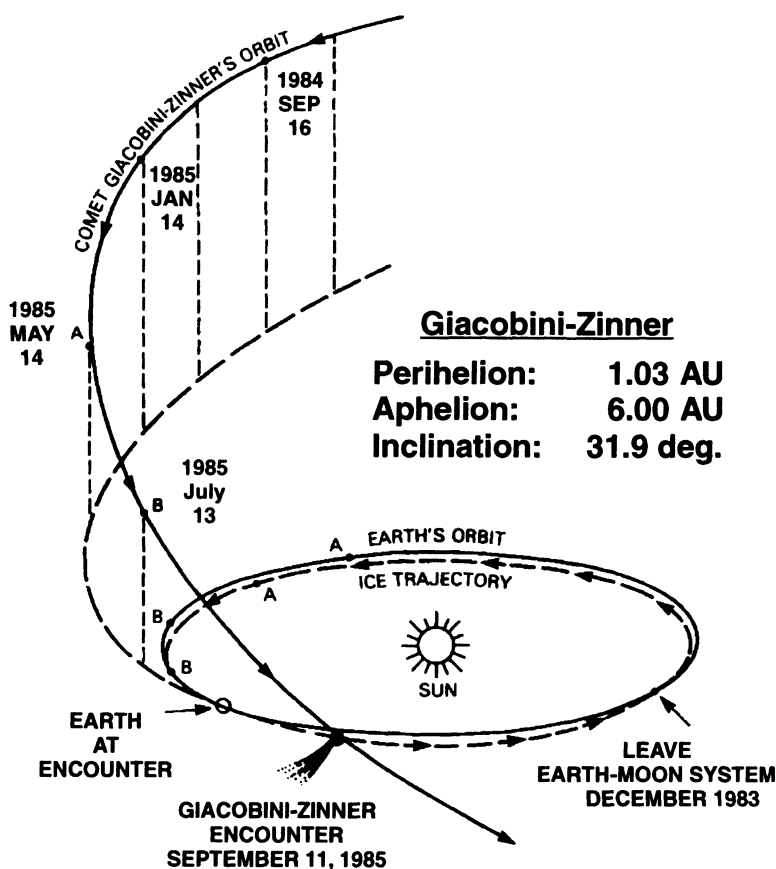


FIG. 25. ICE Heliocentric Trajectory with the Orbits of Earth and Comet Giacobini-Zinner.

Trajectory correction maneuvers were executed on June 5 and July 9, 1985. The first maneuver was about 39 m/sec and the second roughly 1 m/sec for a total ΔV cost of approximately 40 m/sec. In addition to retargeting ICE to the nominal aim point in the comet's tail, these maneuvers also adjusted the timing of the encounter so that it would occur at a time when overlapping coverage from DSN antennas would be possible. As shown in Fig. 27, the tail crossing was centered at 11:00 GMT on September 11, 1985 when three antennas were able to receive data from the ICE spacecraft.

Ground antenna coverage of the ICE comet encounter was quite robust. In addition to arrayed support at the three DSN sites, Madrid, Goldstone, and Canberra, the 64-meter antenna at Usuda, Japan, and the 305-meter antenna of the Arecibo Radio Observatory in Puerto Rico were utilized. A complete discussion of this unprecedented coverage from five major facilities can be found in [53].

Everything was proceeding as planned until just a few weeks before the encounter date when significant changes were observed in the comet's position. These changes were the result of a sharp increase in both the quantity and quality of astrometric observations of Giacobini-Zinner as the comet got closer to the Earth. By September 6, it was estimated that ICE would pass within 8000 km from the comet's nucleus, and would miss the aberrated tail axis by 600 km. To get ICE back on track,

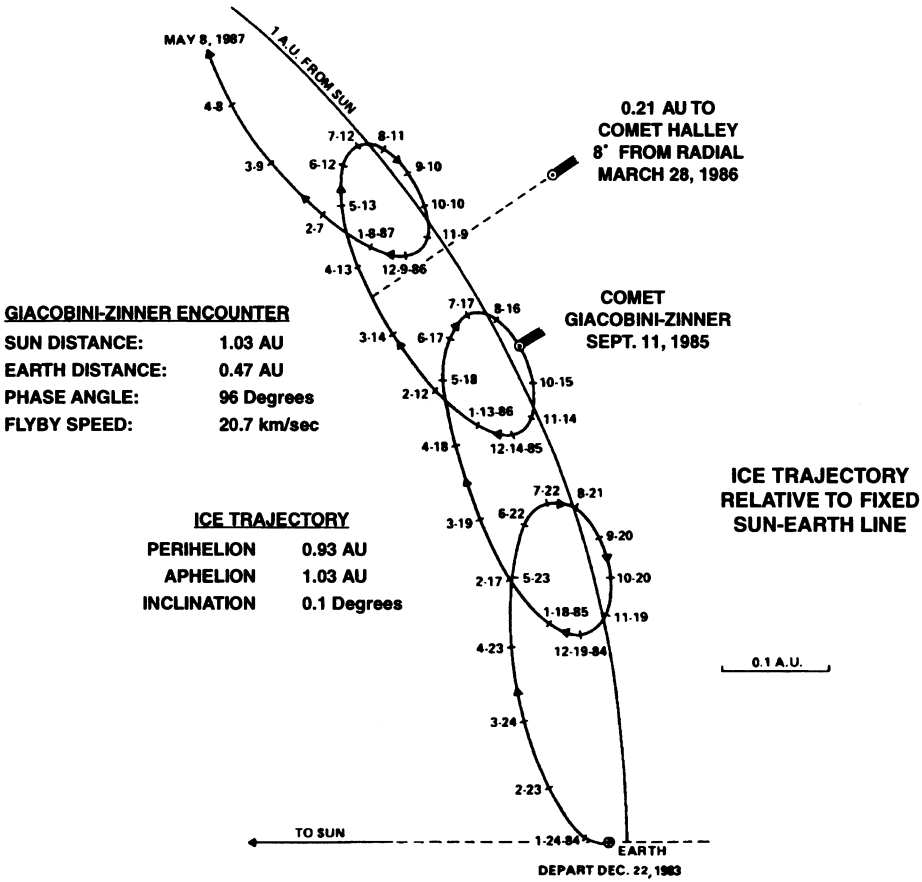
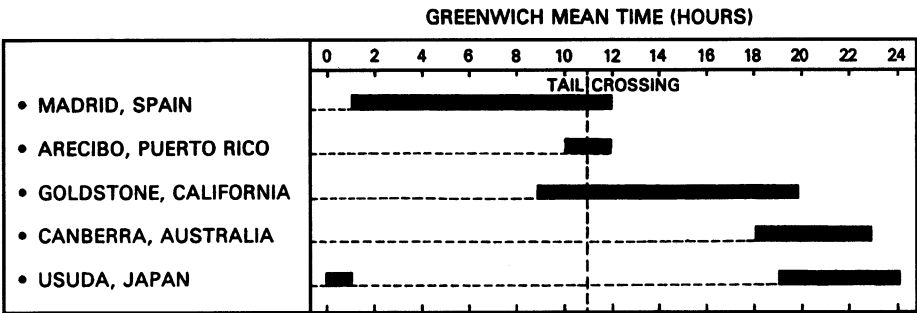


FIG. 26. ICE Heliocentric Trajectory (Bipolar Plot in Ecliptic Plane).



- COMMANDS CAN ONLY BE SENT FROM MADRID OR GOLDSTONE
- REAL-TIME TELEMETRY NOT AVAILABLE FROM CANBERRA OR USUDA

FIG. 27. Antenna Coverage of ICE Comet Encounter.

a small ΔV maneuver (2.3 m/sec) was executed on September 8, just three days before the encounter. This maneuver was designed to correct only the 600-km error in the tail-axis targeting. The much larger ΔV maneuver that was needed to bring the miss distance back to 10,000 km was thought to be too risky, and was not implemented [54].

A schematic diagram of the ICE trajectory through the tail of Giacobini-Zinner is shown in Fig. 28. Although the spacecraft was inside the turbulent interaction region for a little more than three hours (09:10–12:20 UT), the first signs of the comet's presence were noticed in the energetic-ion data about one day before closest approach. The magnetic field structure embedded inside the cold plasma tail was actually seen to be draped as predicted, with two lobes of opposite polarity and—as an indication that ICE had passed very close to the tail axis—the neutral sheet was crossed near the time of closest approach, which occurred at 11:02 UT. Although ICE did not carry a dust experiment, it was possible to deduce information on dust impacts from the plasma-wave data. Luckily, the dust impact rate was much lower than had been feared prior to the encounter.

The rich harvest of scientific results from this first *in-situ* exploration of a cometary environment exceeded all expectations. There were many surprises: the detailed structures in the plasma, magnetic fields, and particle data, the high level of turbulence, the lack of a traditional bow shock, and the detection of “pickup ions” at large distances from the nucleus. Detailed reviews of the scientific results from the ICE comet encounter are given in [55–62].

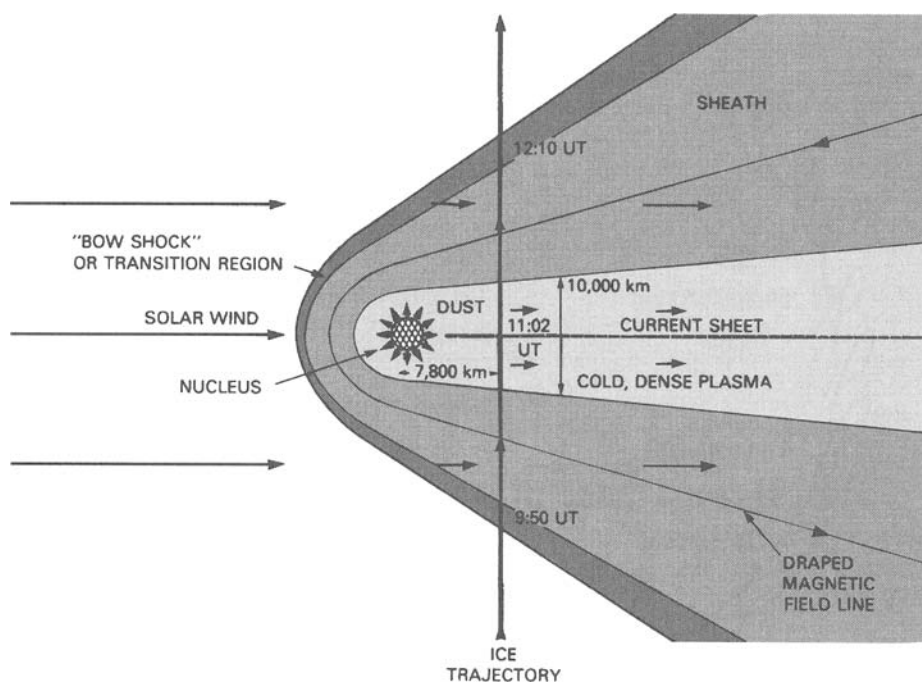


FIG. 28. ICE Trajectory Through Tail of Comet Giacobini-Zinner.

As anticipated, media coverage of the ICE encounter with comet Giacobini-Zinner was extensive. Representative samples of this coverage from both technical and nontechnical publications can be found in [63–72].

Beyond the Giacobini-Zinner Encounter

ICE came through the cometary encounter unscathed, and resumed its boring but useful function of monitoring the interplanetary medium. However, because ICE still had a ΔV capability of 190 m/sec, it was decided to use about 40 m/sec of this capability to return ICE to the Earth-Moon system in 2014. Specifically, it was decided to target ICE for a lunar gravity-assist maneuver on August 10, 2014 that could be used to capture the spacecraft into a loose geocentric orbit [54].

The Earth-return trajectory for ICE is shown in Fig. 29. At the present time, ICE is operating normally, and has not shown any signs of “old age.” As a matter of fact, there have not been any instrument or subsystem failures since 1981. However, because the spacecraft is now 2 AU from the Earth, it has been necessary to reduce its bit rate to 64 bits/sec.

During its solar conjunction period in 1999, ICE will conduct a series of radio-propagation experiments with DSN stations to investigate electron densities in the Sun’s corona. Shortly thereafter, ICE is scheduled to retire from active service, and will be put into a hibernation mode until at least 2010. It is hoped that ICE will be reactivated in 2010, and subsequently return to the vicinity of the Earth where a variety of extended-mission proposals could be implemented.

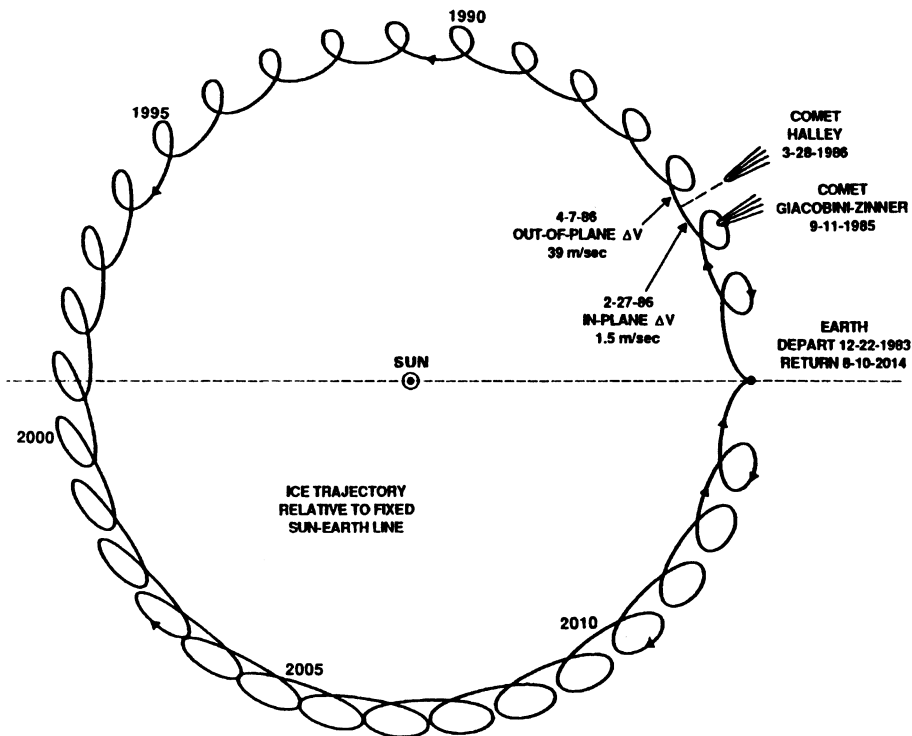


FIG. 29. ICE Earth-Return Trajectory.

Two of these extended-mission options assume that the Moon’s gravity would be used to capture ICE into a high Earth orbit. Once this is accomplished, it would be possible to:

- (1) Return ICE to its original halo orbit around the Sun-Earth L1 libration point. This action would refute the claim of some ISEE-3 investigators that the spacecraft was “stolen” in 1982. It would be shown that the spacecraft was merely “borrowed” for awhile (i.e., thirty-two years!).
- (2) Utilize a series of aerobraking maneuvers to take ICE into a low Earth orbit where it could be retrieved by the Space Shuttle and returned to Earth. Ownership of the spacecraft would then be transferred to the National Air and Space Museum (NASM). A Joint Declaration of Intent to implement this transfer was signed by NASA and NASM on September 11, 1986 (see Appendix D).

Another somewhat more-exciting use for ICE has recently surfaced—that is to use an Earth gravity-assist maneuver to retarget ICE for a second encounter with comet Giacobini-Zinner. The basic plan for this option is outlined in Fig. 30. Notice that this encounter would be even closer to the Earth (0.40 AU) than the previous encounter in 1985 (0.47 AU). An intriguing aspect of the comet option is that another NASA spacecraft called CONTOUR (Comet Nucleus Tour) might intercept Giacobini-Zinner about two weeks later on October 2, 2018 [73]. The instrumentation on CONTOUR is complementary to the ICE instruments, and includes two cameras, a spectral mapper, a dust analyzer, and a neutral gas and ion mass spectrometer.

The Legacy of ISEE-3/ICE

The importance of libration-point orbits and lunar gravity-assist trajectories in the exploration and exploitation of space in the Earth’s neighborhood has become a widely-accepted fact in the 1990’s. However, it was not until the successful flight of

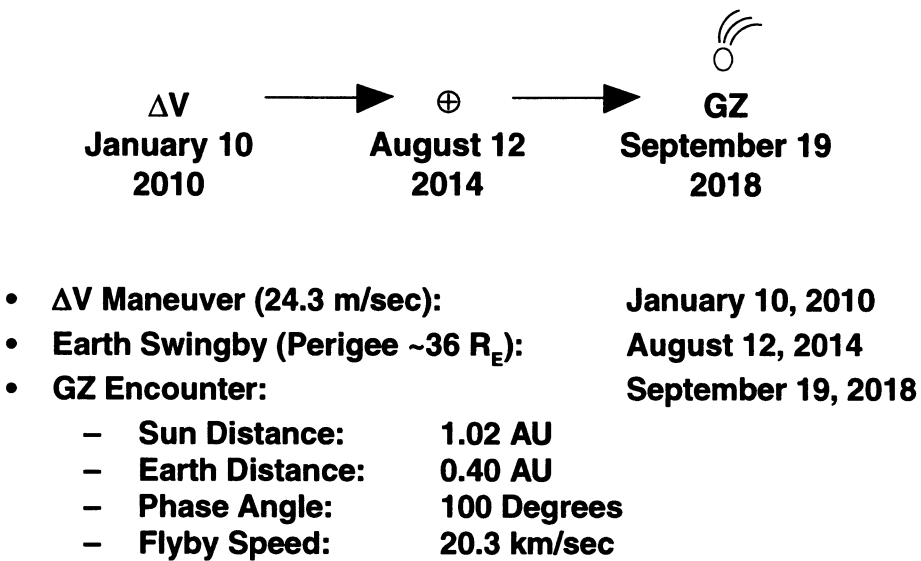


FIG. 30. Return to Comet Giacobini-Zinner?

ISEE-3/ICE that these trajectory techniques gained credibility as practical mission-design approaches. Between 1990 and 1994, three spacecraft have used Double Lunar-Swingby (DLS) trajectories to accomplish their mission objectives. In 1996, and again in 1997, spacecraft have been placed into halo orbits around the Sun-Earth L1 libration point. At least six more libration-point missions are planned to be launched in the next decade. Another DLS mission is also scheduled during this period. All of these missions are a direct result of the success of ISEE-3/ICE and its innovative trajectories.

A listing of past, present, and future libration-point missions is given in Table 4. The gap of eighteen years between ISEE-3 and the second libration-point mission, SOHO, is indicative of the fact that new ideas are not easily accepted.¹³

Four of the missions listed in Table 4, MAP, FIRST, Planck, and NGST, refer to space observatories located in orbits around the Sun-Earth L2 point. This is not a new idea. It was originally put forward by C. A. Cross in 1954 [86]. Although Cross did not provide many details, he was able to see how this location could be advantageous. A recent and more complete discussion of the advantages of Sun-Earth L2 orbits for space observatories was given by R. Farquhar and D. Dunham in [87]. This publication also showed how phasing orbits and a single lunar gravity-assist maneuver could be used to place a spacecraft into a small-amplitude orbit around the Sun-Earth L2 point without an insertion ΔV maneuver. This trajectory, which was originally computed for the now-defunct Russian Relikt-2 mission proposal, is shown in Fig. 31.¹⁴

A listing of missions that have used the DLS trajectory concept is given in Table 5. Two of these missions, Geotail and Wind, belong to a multi-spacecraft entity known as the International Solar-Terrestrial Physics (ISTP) Program [92]. The dynamical heart of this program is the use of the DLS technique that enables the Geotail spacecraft to carry out repeated traverses of the deep-tail region, and the Wind spacecraft to spend 80 to 90% of its time in the solar wind near the Sun-Earth line upstream

TABLE 4. Libration-Point Missions

Mission*	Sun-Earth Libration Point	Date of Orbit Insertion	Mission Purpose	References
ISEE-3 [NASA]	L1, L2	1978, 83	Solar Wind, Cosmic Rays	—
SOHO [ESA/NASA]	L1	1996	Solar Observatory	[74–76]
ACE [NASA]	L1	1997	Solar Wind, Energetic Particles	[77]
MAP [NASA]	L2	2001	Cosmic Microwave Background	[78–79]
Genesis [NASA]	L1	2001	Solar-Wind Composition	[80–81]
Triana [NASA]	L1	2001	Earth Observatory	[82]
FIRST [ESA]	L2	2007	Far Infrared Telescope	[83]
Planck [NASA]	L2	2007	Cosmic Microwave Background	[84]
NGST [NASA]	L2	2007	Deep-Space Observatory	[79, 85]

*Acronyms:

ISEE (International Sun-Earth Explorer)	SOHO (Solar Heliosphere Observatory)
ACE (Advanced Composition Explorer)	MAP (Microwave Anisotropy Probe)
FIRST (Far Infrared Submillimetre Telescope)	NGST (Next Generation Space Telescope)

¹³There are three stages to every unorthodox new idea: first, “it won’t work”; second, “it has no value”; and finally, “we thought of it first” and/or “we knew all about it long ago.”

¹⁴Although this result has been known for some time, and has been published in a number of places [87–90], it is constantly being “rediscovered” by mission planners.

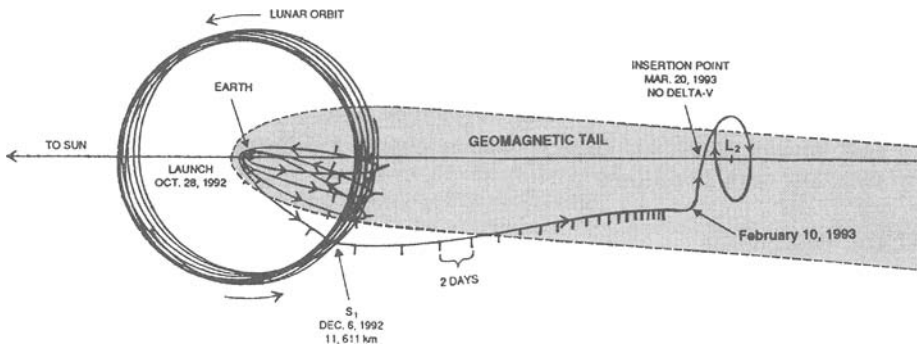


FIG. 31. Relikt-2 Trajectory Using Lunar Gravity-Assist for Transfer to Sun-Earth L2 Point.

TABLE 5. Double Lunar-Swingby Missions

Mission	Date of First Swingby	Purpose	References
ISEE-3 [NASA]	1983	Plasma Studies in Distant Geotail	—
Hiten [Japan]	1990	Engineering Tests	[91]
Geotail [Japan/NASA]	1992	Plasma Studies in Distant Geotail	[92–93]
Wind [NASA]	1994	Solar-Wind Monitor	[92–94]
Magnetospheric Multiscale [NASA]	2005	Multi-Spacecraft Study of Magnetosphere	[96]

from the Earth’s magnetosphere [93–95]. Without the development of the DLS concept, several additional spacecraft would have been required to satisfy the scientific objectives of the ISTP Program.¹⁵

Finally, it should be noted that ISEE-3/ICE is still the only spacecraft to use a lunar gravity-assist maneuver to target a body outside of the Earth-Moon System.¹⁶ However, this situation is expected to change in the near future. Japan’s Planet-B spacecraft is planning to use several lunar swingbys to gain energy on its way to Mars in 1998 [99–100]. Several other planetary and small-body missions, still on drawing boards, are also considering the use of lunar gravity-assist techniques to reach their targets [101–102].

Concluding Remarks

The remarkable accomplishments of ISEE-3/ICE have firmly established its prominent place in spaceflight history. It has distinguished itself in many ways including the following:

- ISEE-3/ICE was the pathfinder for a number of innovative trajectory concepts including halo orbits, double lunar-swingby trajectories, phasing orbits, and escape trajectories using the Moon’s gravity.

¹⁵An attempt to patent the DLS idea was initiated in 1979 (see Appendix E). Unfortunately, the Patent Counsel at NASA’s Goddard Space Flight Center ruled that there was little likelihood that a DLS trajectory would ever be used by NASA.

¹⁶The Clementine spacecraft had planned to use the Moon’s gravity to reach the near-Earth asteroid 1620 Geographos in August 1994, but a major software failure aborted this mission in May 1994 [97–98].

- ISEE-3/ICE has compiled a notable list of scientific achievements including the first exploration of the distant geotail and the first encounter with a comet.
- ISEE-3/ICE has successfully carried out three different missions with a single spacecraft. The cost of the original libration-point mission was approximately \$26 million (real-year dollars). The surcharge for the two extended missions was less than \$2 million!

Acknowledgment

It is a sincere pleasure to thank my close colleagues, David Dunham and Daniel Muhonen, for their insightful advice, and great skill in completing the many computer simulations that were needed to accomplish the ISEE-3/ICE mission.

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
ELECTRONICS RESEARCH CENTER
CAMBRIDGE, MA 02139

PLEASE REPLY TO CODE RT

January 6, 1969

Mr. George M. Low, Code PA
Manager, Apollo Spacecraft Program Office
Manned Spacecraft Center
Houston, Texas 77058

Dear Mr. Low:

The flight of Apollo 8 has focused attention on the absence of a lunar far-side communications capability in the Apollo program. Although two engine burns were successfully performed while the spacecraft was occulted by the Moon, it seems to me that these critical maneuvers would have been less hazardous if they had been monitored by ground stations. (Emergency situations can develop quite rapidly as demonstrated by Gemini VIII). In future Apollo missions, a lunar far-side communications link will be even more desirable since other critical operations such as rendezvous, docking, and extravehicular activity might also take place during occultation periods.

The lunar far-side communications problem has concerned me for some time, and I believe that it can best be solved by placing a single relay satellite in a "halo orbit" that would be visible from the occulted spacecraft as well as the Earth. Placing a relay satellite in the halo orbit should not be too difficult, and the ΔV penalty is actually smaller than that which is needed to attain a synchronous Earth orbit. Although some station-keeping would be required to maintain the relay satellite in this halo orbit, the fuel consumption would not exceed the capacity of existing satellite translation-control systems. A detailed discussion of the halo-orbit communications concept is contained in the enclosures.

It is not my desire to introduce an unwelcome communications constraint into the Apollo program at this time. However, the need for lunar far-side communications will be even more apparent in the post-Apollo period. I would be happy to discuss the advantages of this halo orbit concept in more detail with someone

In the appropriate Program Office. Any assistance that you could render in putting me in touch with the proper people would be greatly appreciated.

Sincerely,

Robert W. Farquhar

Robert W. Farquhar
Office of Control Theory
and Application

Enclosures: 3

1. "Libration-Point Satellites: Their Role in Future Space Missions"
2. "Lunar Communications with Libration-Point Satellites"
3. "The Control and Use of Libration-Point Satellites"

cc:

Mr. O. Hugo Schuck
Director, Office of Control Theory and Application
NASA/ERC

Mr. George K. Megerian
Inter-Center Technology Assistance Office
NASA/ERC



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
MANNED SPACECRAFT CENTER
HOUSTON, TEXAS 77058

IN REPLY REFER TO: PA-9-1-17

January 10, 1969

Mr. Robert W. Farquhar
Office of Control Theory and Application
NASA Electronics Research Center
Cambridge, Massachusetts 02139

Dear Mr. Farquhar:

Thank you for sending me your recent letter and the material on satellites for lunar far-side communications.

I have forwarded this information to our Director of Flight Operations, Mr. Christopher C. Kraft, who would be responsible at this Center for generating requirements for lunar communications capabilities.

Sincerely yours,

George M. Low

George M. Low
Manager
Apollo Spacecraft Program

MEETING MINUTES
MSC DRS STEERING COMMITTEE MEETING

January 7, 1970

A. ATTENDEES

The subject meeting was held at 1:00 p.m. in Room 3007, Building 30.

Members present were:

Sharon Henderson - AMFC/HD
Vernon M. Dauphin - SAAD/TE
Robert W. Moorehead - RAD/EE8
David T. Myles, Jr. - POD/FSL

David R. Brooks (POD), the only member absent was at KSC on official business. Other attendees invited specifically by the Steering Committee were:

George D. Arndt - RAD/EE8
Sidney W. Kovacs - RAD/EE8
James A. Porter - RAD/EE8
J. Gorman - FHO/EE8
George Q. Clark - GSCC/Code 811
John W. Corbman - GSCC/Code 811
Robert D. Goffrey - GSCC/Code 831
Robert W. Farquhar - GSCC/Code 551

B. LUNAR RELAY STUDIES

The meeting was devoted primarily to the subject of lunar data relay satellites with both MSC and GSCC presenting study plans.

Mr. Arndt presented an outline from a list of trade studies dealing with lunar relays which RAD is performing inhouse and expects to complete by mid-March. He also presented a list of "technology" studies dealing with space communications which RAD plans to undertake in the near and distant future. Mr. Moorehead pointed out that any of his study efforts dealing with lunar relay satellites are low priority since the emphasis in RAD is currently on the space shuttle system.

Mr. Clark then gave a very interesting presentation proposing a course of action to provide far-side lunar communications for supporting lunar far-side exploration. The Apollo Lunar Exploration Office at NASA Headquarters had asked GSCC to perform a feasibility study (about \$100K worth) dealing with this problem. However, GSCC (Tracking and Data Systems Directorate and Manned Flight Support Directorate) felt that enough feasibility work had already been done, and that the proper course of action would be to initiate a combined Phase A/B effort (about \$700K worth) if a requirement really existed. This presentation by Mr. Clark was to be given later this month to the Apollo Lunar Exploration Office; however, Mr. Farquhar received a telephone call prior to this Steering Committee Meeting informing him that the study money is no longer available. The major points of interest in Mr. Clark's presentation were:

- a. GSCC, having performed numerous feasibility trade studies, believes that a communication relay satellite in a Halo orbit (about the L-2 libration point) provides the optimum placement solution for a lunar relay in the mid 1970's.
- b. Feasibility studies have been performed by GSCC showing that this relay could provide total Apollo communications capability from the lunar far-side to earth.
- c. A project to provide this lunar relay would cost on the order of \$70 million.
- d. The projected lifetime of the relay satellite would be 3 years.
- e. This particular type of relay satellite could be operational by late 1973 or early 1974.

Dr. J. F. Clark - Director

September 7, 1971

Dr. R. W. Farquhar
Flight Mission Analysis Branch

Halo-Orbit Comsat for Apollo 17

The possibility of performing a far-side lunar landing in the crater Tsolokovsky with Apollo 17 is being considered by the Apollo Spacecraft Program Office of MSC. Preliminary investigations have quickly led to the conclusion that a halo-orbit data-relay satellite would be mandatory for such a mission. Therefore, the undersigned was invited to participate in meetings at MSC concerning the feasibility of placing an "off-the-shelf" comsat into a lunar halo orbit by December 1972.

On September 2, 1971, the results of a brief study of the proposed Tsolokovsky mission were presented to Col. James A. McDivitt, Apollo 17 crew members, and management from other MSC Directorates. The overall concept of using a halo-orbit comsat for lunar far-side communications support as well as details of trajectory and stationkeeping requirements for halo satellites were presented by the undersigned. Apollo interface problems and candidate spacecraft hardware concepts were discussed by MSC personnel. The main conclusions were:

1. The Tsolokovsky mission is technically feasible.
2. A suitable halo-orbit comsat could be obtained by using slightly modified versions of either the Hughes Intelsat-IV or the TRW BSCS-II. It would be possible to have two spacecraft ready for launch by November 1972 (see attached schedule).
3. The additional costs that would be incurred (see attachment) would require a substantial agency commitment.

After the presentation, Col. McDivitt said that he would bring this matter to the attention of NASA Headquarters as soon as possible since a decision would be needed rather soon.

2

Subject: Halo-Orbit Comsat for Apollo 17

Meanwhile, the in-house studies at MSC are continuing, and GSFC is providing MSC with halo-satellite data as needed.

Original Signed by

Robert W. Farquhar
Flight Mission Analysis Branch
Trajectory Analysis & Geodynamics Division
Mission and Data Operations Directorate

Attachment

cc: Mr. J. T. Mangel, Code 500, w/attach
Dr. F. O. Vonbun, Code 550, w/attach
Mr. E. T. Groves, Code 551, w/attach

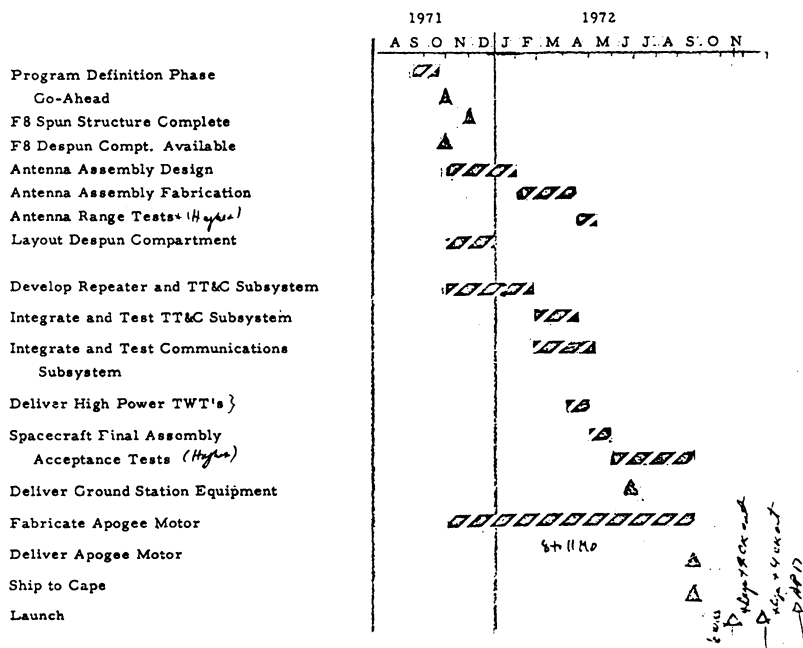
551:313M:RWF:rjs

Document A-3 Memo from Robert Farquhar to John Clark,
Director of Goddard Space Flight Center, dated September 7, 1971.

COMMUNICATIONS RELAY PRELIMINARY COST ESTIMATES

	<u>DSCS - II</u>	<u>INTELSAT IV</u>
SPACECRAFT	20 M	20 M
SECOND SPACECRAFT	6 M	12 M
ATLAS/CENTAUR		15 M
SECOND ATLAS		15 M
TITAN III C	22 M	
APOLLO MODIFICATIONS	15 M	15 M
TOTAL	63 M	77 M
NO SECOND LAUNCH		- 20 M
REVISED TOTAL		57 M

PROGRAM SCHEDULE FOR RELAY SATELLITE



Halo Effect Developed

Way To Communicate With Far Side Of Moon Found

Greenbelt, Md. (AP)—A scientist at the Goddard Space Flight Center says he has discovered a way for astronauts to remain in continuous communication with earth while landing on the far side of the moon.

The key to the technique would be a communications satellite orbiting an invisible point: 40,000 miles beyond the moon, says Robert W. Farquhar, the engineer-mathematician who developed the idea.

Appears As Halo

By orbiting that point, rather than the moon itself, the satellite could keep watch on any desired location on the back side of the moon while maintaining constant communication with earth, Farquhar says.

From earth, the satellite would appear to describe a sort of halo around the moon, prompting Farquhar to label it the halo satellite.

The halo satellite would orbit L-2, one of five libration points where the gravitational pull of the earth and moon are equal.

In theory, a spacecraft placed on one of the libration points would hover there indefinitely, delicately balanced between the pull of the earth and moon.

However, the libration points oscillate and this instability means that, in practice, a satellite orbiting one of them would need a small amount of fuel to maintain its required position.

Vetted By NASA

Farquhar says he presented the halo-satellite idea to space agency officials in time for it to

be used on the Apollo 17 space flight scheduled for December. He said that budget-conscious officials at NASA vetoed the idea because of \$50 million added cost.

"We should have done it because there is no other lunar program in sight for 15 or 20 years," Farquhar said.

The National Aeronautics and Space Administration has no manned moon landings scheduled after Apollo 17.

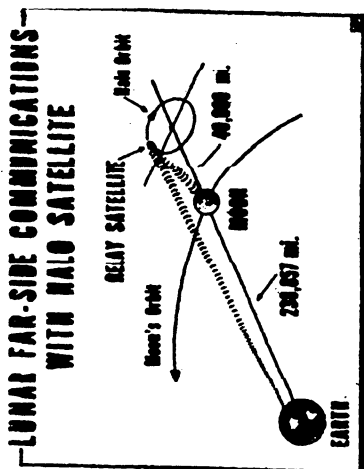
Future Consideration

Dr. Richard B. Allenby, assistant director for lunar science in NASA's Office of Manned Space Flight, said the agency will consider Farquhar's proposal for any future manned moon landings that might be scheduled later.

"It (Farquhar) has a very good idea," Allenby said. "His concept is very sound and we are very interested."

Lunar scientists are eager to discover what the far side of the moon has the same general composition as the front side. They are particularly curious about the absence of maria, or dry seas, which are so numerous on the front side.

However, space officials have been reluctant to attempt a far-side landing without some means of continuous communication between earth and the crew making the landing.



'HALO' SATELLITE—Robert W. Farquhar, a scientist, says that the use of a communications satellite orbiting an invisible point more than 40,000 miles beyond the moon would enable astronauts landing on the far side of the moon to stay in contact with earth headquarters.



National Aeronautics and
Space Administration

Washington, D.C.
20546

Reply to Attn of **SL-4**

APR 27 1981

Mr. A. Thomas Young
Director
Goddard Space Flight Center
Greenbelt, MD 20771

Dear Tom:

I have been made aware of the possibility of deflecting the ISEE spacecraft to intercept comet Halley. Since we are continually assessing what our total involvement should be with respect to the apparition of Halley's comet, it behooves us to understand all of the options available to us. While I am, in no way, endorsing or supporting the potential of an ISEE intercept with comet Halley, I would like to have the information as to whether the mission is even physically realizable. On this basis, I would support a small study at GSFC, to be led by Dr. Farquhar, to evaluate the feasibility of the mission he proposed. Dr. Farquhar should understand that our desire is solely to gain information, and he should not represent the authority to conduct the study as an endorsement of this mission by the Office of Space Science.

This study will be funded and managed by the Advanced Programs area of the Solar System Exploration Division, and would, therefore, be under the direct cognizance of Mr. Diaz.

Sincerely,

A handwritten signature in dark ink, appearing to read "Andrew J. Stofan".

Andrew J. Stofan
Acting Associate Administrator
for Space Science

Document B-1 Memo from Andrew Stofan to A. Thomas Young, dated April 27, 1981.

602

TO: NASA Headquarters
Attention: Office of Space Science
Dr. Erwin R. Schmerling/SC-7

NASA Headquarters
Attention: Office of Space Science
Mr. Herbert B. Chisholm/SC-7

FROM: Sciences Directorate
Orbiting Satellites Project/Science

SUBJECT: Move of ISEE-3 to the Geomagnetic Tail

At the recent ISEE SWT meeting in France, a majority of the members present requested that the Project investigate the feasibility of moving ISEE-3 to the geomagnetic tail. Further, it was requested that the results of this study be presented at the next SWT meeting to be held at the GSFC in February 1982.

Dr. Tycho Von Rosenvinge has discussed the scientific aspects of the move with Bob Farquhar, who would head up the study. The enclosure describes the study which is necessary to satisfy the request of the SWT. To accomplish this study, an additional \$50K over the present ISEE guideline is required at the start of FY82. The dollar requirements for actually implementing the maneuver will be identified during the study.

If the Program Office desires that the study be conducted, please inform the Project Office and make the appropriate arrangement for the additional \$50K.

ORIGINAL SIGNED BY

J. P. Corrigan

Enclosure

cc: R. Farquhar/580
J. Cooley/581.3
P. Corrigan/602
R. Wales/602
T. Von Rosenvinge/661
100
600 w/o encl
602
Mailroom/251.2

602/JPC/ss:8/25/81

**Document B-2 Memo from J. Patrick Corrigan to Erwin Schmerling,
dated August 25, 1981.**

DEC 21 1981

EZ-7

TO: Goddard Space Flight Center
 Attn: Code 620/Dr. George F. Pieper
 Code 622/Mr. J. Patrick Corrigan

FROM: EZ-7/Director, Astrophysics Division

SUBJECT: Feasibility Studies of Extended Mission Options
 for ISEE-3

REF: Memo from J. J. Corrigan to E. R. Schmerling, dtd
 August 25, 1981. Subject: Move of ISEE-3 to the
 Geomagnetic Tail

As you already know, I have received a request to authorize a feasibility study for a transfer of the ISEE-3 spacecraft from its present halo orbit to the geomagnetic-tail region. Since receiving this request, I have learned that it might also be possible to send ISEE-3 towards an encounter with the comet, Giacobini-Zinner, after the geomagnetic-tail excursion. The comet option has elicited some interest from Mr. D. Herman and Dr. G. Briggs of the Earth and Planetary Exploration Division.

Although we would not endorse or support an ISEE-3 comet intercept at this time, we would like to have information concerning the physical possibility of this extended mission. Therefore, I would like to have the comet option included in the feasibility studies of extended mission alternatives for ISEE-3.

Please submit a revised plan for the ISEE-3 feasibility study. The revised study plan should include two mission options following the move of ISEE-3 to the geomagnetic-tail. In one option, ISEE-3 would return to a halo orbit upstream of the magnetosphere after spending a year or more in the tail. In the other option, the spacecraft would be diverted to intercept comet, Giacobini-Zinner, instead of returning to the halo-orbit. When submitting the new study plan, please include the funding requirements that will be needed to carry out this investigation.

I would like to emphasize that you are authorized to study options only. I have no intention of proceeding with anything other than the nominal ISEE-3 operation until the options are thoroughly discussed with the appropriate members of the science community and the ISEE-3 science team. This must be done within the framework of the existing ISEE budget.

Very truly yours,

Franklin D. Martin

cc:
 EL-4/Moore
 Herman
 Briggs
 ZE-2/Tilford
 Schmerling
 EZ-7/Halpern
 GSFC/Farquhar ✓
 Quann
 JPL/Albee
 EZ-7REHalpern:ep/12-14-81/53657

Document B-3 Memo from Franklin Martin to George Pieper, dated December 21, 1981.



University of Hawaii at Manoa

Institute for Astronomy
2880 Woodlawn Drive • Honolulu, Hawaii 96822
Telex: 723-8459 • UHAST HR

May 14, 1982

Dr. Burton J. Edelson
Code E
NASA Headquarters
Washington, D.C. 20546

Dear Burt:

I am writing on behalf of the Comet and Asteroid Working Group of the Solar System Exploration Committee (SSEC) to support the ISEE-3 opportunity to explore the plasma environment of Comet Giacobini-Zinner.

Our Working Group has been examining a variety of options for initiating the exploration of the small (and chemically primitive) members of the solar system, and we are enthusiastic about both the scientific and exploratory value of these objects and the number of favorable launch opportunities coming up in the late 1980's and early 1990's. I am convinced we can and will meet this exciting challenge to open for study the third major element of the planetary system. Realistically, however, we will not be able to acquire these data for another decade, so the unique opportunity to utilize an existing spacecraft for a first pass by a comet is particularly appealing. Although the ISEE-3 instrumentation will not permit direct investigation of the nucleus or neutral coma of the comet, it will be able to provide valuable data on the interaction of the comet with the solar wind, data never before obtained for any comet. It also permits the early return of these results--even before the USSR, ESA, and Japanese spacecraft encounter Halley. This use of the ISEE-3 spacecraft in its extended mission mode has our warm endorsement, and it is hard to imagine that NASA would decline this opportunity to obtain new and unique data with such a small expenditure of resources.

I am pleased to have spoken to you briefly a week ago in Washington, and I look forward to remaining in touch.

Sincerely,

David Morrison, Chairman,
SSEC Working Group on Comet
and Asteroid Missions

DM:dr

xc: Charles Pellerin
Geoff Briggs
Noel Hinners

AN EQUAL OPPORTUNITY EMPLOYER

National Aeronautics and
Space Administration
Goldard Space Flight Center
Greenbelt, Maryland
20771



402

TO: NASA Headquarters
Attention: Office of Space Science & Applications
Dr. Franklin D. Martin/EZ-7

FROM: Director of Sciences

SUBJECT: International Sun-Earth Explorer-3 (ISEE-3)

A small but significant orbit maneuver for ISEE-3 is planned for June 10, 1982. The SEE-3 is, as you know, in a halo-type orbit in the vicinity of the sunward libration point, L₁, of the Sun-Earth system. The spacecraft has periodically required station keeping maneuvers to keep its orbit properly aligned with the Sun and Earth to allow the gathering of scientific data. For this particular maneuver I have authorized the maneuvering of the spacecraft in such a way as to keep open several extended mission options which are presently being actively investigated.

The forthcoming maneuver which consists of adding a velocity increment of 4.6 meters per second will, if not rescinded, cause the spacecraft to start a trajectory which will carry it toward and subsequently into the geomagnetic tail. The enclosed sketch (Figure 1) shows the orbital path.

It is important that this maneuver be done at this time because it keeps open the opportunities for the following extended mission options:

- geomagnetic tail excursion followed by return to halo orbit,
- geomagnetic tail excursion and remain in the tail,
- geomagnetic tail excursion followed by an encounter with the comet Giacobini-Zinner, and
- remain in halo orbit.

The key features of the timing of the maneuver are subsequent necessary lunar flybys and perigees when the spacecraft can use the Earth and Moon for trajectory modification for either a return to the halo orbit at L₁ or the start of a trajectory which would lead to the cometary encounter (options a. and c. above).

The cost, in onboard propellant, of rescinding the planned mid-course maneuver increases with time to the beginning of August after which it would be too costly to stop the spacecraft from going on a trajectory to the geomagnetic tail. Figure 2 shows the velocity increment required to rescind the planned maneuver increases from 4.6 meters per second to about 130 meters per second by mid-August. The present total maneuver capability of ISEE-3 is about 350 meters per second. The cost per year of remaining in the geomagnetic tail orbit is between 10 and 100 meters per second depending on the kinds of orbit selected. The risk associated with going to the geomagnetic tail is about the same as leaving the spacecraft in its present orbit. In either case, the propulsion system will have to be used. There are no significant ground system requirements for the geomagnetic tail excursion. Everything would be about the same except that the spacecraft would be in an interesting and unexplored region of space.

From a physical point of view, the decision concerning the comet encounter does not need to be made until April 1983. Budgetary planning will force it to be made this summer, I believe.

Unless you advise me to the contrary, the June 10, 1982, orbit maneuver will be carried out.

George F. Pieper
George F. Pieper

Enclosures: 2
cc: C. Peiler/IN/HQE

NASA Form 1187
(Nov. 66)

• GPO: 1966-O-335-961

National Aeronautics and
Space Administration
Washington, D. C.
20546

EXECUTIVE CORRESPONDENCE

*Boo - Advance copy of the
approval letter. This has
not been sent as of 9/9/82*

AUG 30 1982

Reply to Attn of: EZ-7

TO: Goddard Space Flight Center
Attn: 100/Director

FROM: S/Associate Administrator for Space Science and
Applications

SUBJECT: ISEE-2

The mission of ISEE-2 to explore the comet Giacobini-Zinner is approved. This approval is based on the presentation and discussion of August 6, 1982, and the draft Project Plan provided by GSFC. If there are major changes in the technical risk, or in the funding requirements for either OSCA or OSTDS, please assure that I am promptly informed.

B. I. Edelson
B. I. Edelson

Document B-6 Executive Correspondence from Burton Edelson to the Director,
Goddard Space Flight Center (Noel Hinners), dated August 30, 1982.

Giacobini-Zinner Targeted

U.S. Satellite Will Be First to Reach a Comet

By LEE DYE, Times Science Writer

Six months before the first foreign space probes reach Halley's comet, an American spacecraft will pass through the tail of another comet in a snappy demonstration of scientific and bureaucratic derring-do.

The craft is an aging but dependable satellite launched in 1978 and recycled for the new mission over some strong objections within the National Aeronautics and Space Administration.

The mission to comet Giacobini-Zinner falls far short of the dreams of some scientists to send a probe to Halley, the most famous comet of all, but it does mean that the first spacecraft to reach a comet will be from the United States, not the Soviet Union, Japan or Western Europe.

Stationary Orbit

"We wanted to be first," said Robert Farquhar of NASA's Goddard Space Flight Center. He is one of three scientists credited with pulling off the adventure and the one who figured out how to get the craft out of its stationary orbit, rocket it around the moon and on its way to a rendezvous this September with the comet.

Some within NASA are furious over the fact that the International Sun and Earth Explorer was diverted from other duties and sent off to chase a comet, but Farquhar and his cohorts show no shame.

"We stole it," said Fred Scarf, a top scientist at TRW and one of the leaders of the rebellion.

The road to Giacobini-Zinner proved a little bumpier than Scarf had anticipated when he first tried to persuade NASA that Farquhar

could make the moon do what the satellite's small rockets couldn't do on their own—sling the satellite out to the comet. "What Scarf didn't know until after NASA began to show real interest was that Farquhar wasn't sure how to do it."

"Fred set me up," Farquhar said. As it turned out, he set up the right man. Farquhar's calculations have put the craft, renamed the International Cometary Explorer, right on course. About the only thing that remains to be seen at this point is whether the spacecraft will survive a treacherous high-speed flight Sept. 11 through the dust storm that follows the comet.

If it does, and Scarf thinks it might, it will continue on around the sun and return to the vicinity of

Please call (714) 948-1413.

PROBE: American Satellite to Score Space First by Meeting Comet Tail

Continued from Page 1

Earth in 1981. And if that came to pass, Parquhar wants to go out and get it.

The spacecraft was sent up seven years ago, loaded with instruments to study the sun. It came to rest in an area where the gravitational pull of Earth and the sun kept it in a kind of suspended state. It was designed to study solar disturbances that could affect Earth's weather and studying a wide range of solar phenomena. Meanwhile, in 1981, NASA scientists were juggling a plan to send a small spacecraft to study Halley's comet, the first of the U.S. prehistoric space exploration.

"The scientists involved with planning for Halley were a little too ambitious," Parquhar said. "They were more interested in the scientific value of their technological capabilities than in the scientific value of Halley's comet. And they were coming in with projects that would cost almost \$1 billion."

To win support for their proposal, the Halley planners argued against a simple comet fly-by, but they argued for a more expensive, NASA abandoned the idea of going to the famous comet.

Japan, a consortium of Western European nations and the Soviet Union all announced missions to Halley, meaning that the United States would be the only nation with a space probe to the comet. But it was not sending a probe to the comet.

This disappointment weighed heavily on many members of the scientific community, and in August 1982, Parquhar and Dr. Laboratory palmer, with other scientists at Goddard to discuss the future of the International Sun and

Earth Explorer, as it was known then. "The scientists, who had experience aboard the craft, were members of an advisory committee set up by NASA on the use of the spacecraft."

The three scientists were concerned over the wisdom of leaving the craft where it was, doing the same thing it had already done for seven years. If there was something about it that it could do—like intercept a comet.

Parquhar told the down or to scientists that the craft could be sent around behind the Earth, where it could study materials that trail the planet as it revolves around the sun, giving the craftsman a view of the comet's tail.

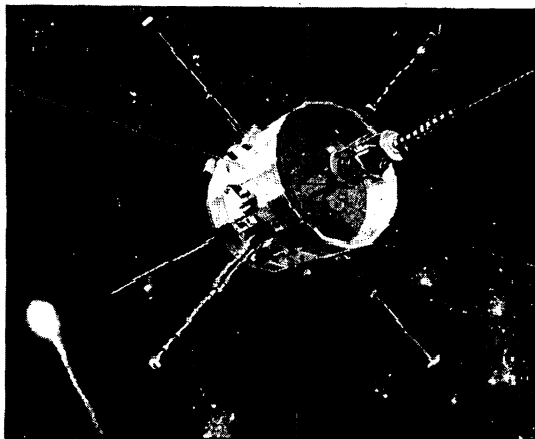
Everybody seemed to like the idea. Then he dropped it bombshell. "Sear said," Parquhar has figured out how to get the spacecraft out to intercept Giacobini-Zinner, "an intercept mission that passes the Earth every 64 years."

"What I didn't know at that time was that Bob at that point, didn't know how to get us there," Sear said.

Parquhar agrees. "I had already fixed on the side about this comet thing, but I didn't have all the details worked out yet, and I really wasn't very sure it could be done," he said.

Parquhar said he was kind of "in the spot," and he did not tell the advisory committee that it did not know how to do it.

The committee voted overwhelmingly against it, however, because most of the instruments aboard the craft were designed to study the sun, no comets, and would not be useful in an intercept mission with Giacobini-Zinner. Smith, who said he befuddled the idea, was put in charge of a



Artist's conception shows U.S. spacecraft's encounter with the Giacobini-Zinner comet, scheduled to occur on Sept. 11.

committee to take the matter before the National Academy of Sciences for an independent assessment. Parquhar and Smith were on the committee, but the recommendation of the advisory committee was rejected, something NASA rarely does.

The contrary race was on.

"I kind of got caught up in the spirit," said Parquhar, who still believes the mission was a kind of "knee" I was going to look kind of stupid if it didn't work. We tried several things and none of them worked," because, he said, the moon

and the Earth were not in the right position relative to each other.

"The phasing is very difficult," he added. "In order to get out to the comet, I had to pass the moon at just the right angle at just the right time just to be right again."

The idea was to pass the moon close enough to the Earth to whip the craft out toward the comet. Using sophisticated computers and models of the solar system, Parquhar finally determined that it would work if he fired the spacecraft's rocket several times, "one at a time," he said.

The spacecraft's rocket motors, built by TRW, were not designed as primary propulsion engines. They were designed to keep the craft on station as it studied the sun, and they are much smaller than the main engines of a rocket. Still, they proved sufficient for the task. In January of last year, the spacecraft rounded the moon and headed out for its encounter with Giacobini-Zinner, and it was given its new name.

Seven instruments aboard the spacecraft are ideally suited to study the comet, among them are Sear's space plasma equipment, which should tell scientists more about the nature of a comet's tail, and Smith's magnetometer, which will study the comet's magnetic field.

But the craft does not carry the full range of equipment found aboard foreign vehicles that were designed specifically for cometary exploration. There are no cameras aboard, for example, so there will be no pictures of the spacecraft of Giacobini-Zinner.

"That does not worry people like Sear because it means the craft does not have to remain on the

sunny side of the comet and is free to fly through its tail mass, gathering information that will be denied to the Halley probe.

What does worry just about everybody is the lack of protection for the spacecraft's instruments. The European Halley probe, which launched last month, is protected with a double shield of the same material used to make bulletproof vests.

When the American craft reaches Giacobini-Zinner, it should be about 100 miles from the comet's tail, but the craft has several antennas that stretch out in several directions nearly the length of a football field. If those are sheared off, most of the data will be lost.

Giacobini-Zinner's tail appears to be made of very loose material. Twice in this century the Earth has passed through the comet's tail, and each time the comet "gave us some of the most intense meteor shower in history," Parquhar said. But particles could damage the craft's instruments. The spacecraft could simply bump its solar antenna no longer points toward Earth.

"I think there is a strong possibility that we may not take it through the encounter," Parquhar said. "I don't think there's a 50-50 chance of a second that doesn't have a shield on it. This is a very risky."

Parquhar said he hopes "we will get data at least halfway through the tail. The craft should eventually miss the comet, but it will be nearly 10 million miles away at its closest approach."

By then, it should be a veteran of the first man-made instrument to have visited a comet.

Doodling the route into a comet's tail

Goddard scientist plotted space craft's encounter

By David L. Chandler
Globe Staff

GREENBELT, Md. — Four years ago, when budget cutbacks killed a \$500 million proposal for the United States to send a space probe to intercept Halley's comet, the prestige of being first to study a comet close-up seemed destined to go to Japan, the USSR, or the European space agency — all of whom were proceeding with probes of their own.

But then Robert Farquhar started thinking about it. And last week, his mathematical doodles culminated in a stunning rendezvous with the little-known comet Giacobini-Zinner.

Not only had he figured out how to save face and collect some important scientific data on comets, but it was all accomplished for less than a hundredth of the cost of the cancelled Halley probe.

His plan worked so well on Wednesday that it has already produced enough surprises to rewrite the textbooks on comets. But it took some doing.

Farquhar, among other things, is an astrodynamical — someone who figures out orbital paths for objects whirling through space. It's a job he's been doing since shortly after the launch of Sputnik in 1957 astonished the world and, incidentally, initiated Farquhar's obsession with getting objects into space and then having them go where you want them to.

Since 1969, he's been working here at Goddard Space Flight Center, a National Aeronautics and Space Administration facility where many deep-space missions are planned and controlled.

Now, the 53-year-old Farquhar may well have achieved the ultimate in his esoteric profession, the space science equivalent of pitching a perfect game in the world series. The plan he devised for man's first comet encounter turned out to be, he now says, "the most complicated thing I've ever been done. I think, in the way of orbital dynamics or moving a spacecraft around."

There was an object already in space, a US satellite launched in 1978 called the International Sun-

COMET, Page 43

Doodling the route to a comet's tail

COMET

Continued from Page 41

Earth Explorer (ISEE) that carried several European experiments. Farquhar had helped to plan and design it, and now he began wondering whether he could get it to go where he wanted it to — to Halley's comet.

It was a tricky problem, because the craft's only power source was a group of tiny maneuvering engines intended only for making minor adjustments in its position. But after a little thought, he figured it could be done.

The trajectory he had in mind made sense, but there turned out to be a fatal problem: ISEE's feeble antenna, designed for use in a near-Earth orbit, wouldn't be able to send any intelligible data back from the distant point where it would finally have met Halley.

So Farquhar thought some more, and drew some more curves and figure eight on his notepad. Halley's was out, but there was another comet — that could be reached while it was much closer. While Halley's is more famous and more visible, all comets are believed to have similar structure and composition so scientists could learn just as much, and reap the same bountiful harvest of data, from one as another.

Comets are common. Several new ones are found every year, and dozens return regularly on known pathways. But they come cloaked in a bright nimbus of gas and dust, and so have preserved their aura of mystery despite recorded observations that go back for millennia.

A chaotic storm

The theoretical picture scientists have drawn of comets holds that there's a tiny nucleus of ice and dust, only a few miles across, that starts to melt as it approaches the sun's heat. This produces a cloud of gas and dust that envelops the nucleus and streams out behind it, sometimes for millions of miles.

The overall picture remains intact after Wednesday's mission, but the details are much more complex and active than expected. The region inside the comet turns out to be a chaotic storm of magnetic fields and electrically charged gases, shot through with high-energy particles traveling at high speed. And the bow shock, the wake the comet makes as it plows through the solar wind, turns out to be surprisingly elusive.

Goddard scientist John Brandt said Friday that the new findings will have "a devastating effect ... now we are going to have to face reality in our modeling."

Farquhar first mentioned the idea of sending ISEE out to meet the little-known comet to his colleague Frederick Scarf, a physicist at TRW Corp. in California, who



Dr. Robert Farquhar and his far-away target, Giacobini-Zinner.

had designed one of the ISEE experiments. Scarf liked the idea, and between them they persuaded NASA to approve the concept.

Then came the hard part. Having decided after a quick analysis that the mission was possible, all of a sudden Farquhar was faced with figuring out how to do it.

"I figured it could be done," Farquhar said in an interview last week. Speaking as he often does, as if he were the spacecraft, he continued, "but I was kind of casual about how I was going to get from the orbit I was in into this orbit to go out to the comet."

Calculating orbits for spacecraft, he explains, is "more of an art than it is just a pure science. It's a little bit of both. I like something where I have to do some of the thinking, not the computer all the time. I don't like computers anyway."

A dislike for computers may be a little odd for an astronautical engineer, but Farquhar has a tendency to do things his own way. And often, others end up doing things his way as well.

He is the inventor of a variety of orbital maneuvers that have since become standard parts of the repertoire of space planners, including something called a "halo orbit" — used in the first phase of the ISEE mission — where a spacecraft orbits around an imaginary point in space, called a libration point, where the gravity of the Earth and the sun exactly balance each other. This orbit, which was the subject of his Ph.D. thesis, provides a stable path where a satellite could continue circling indefinitely just as the moon circles the Earth.

One of his major duties at Goddard is to think up plans for new missions for NASA to take on. Usually, he works out the feasibility of a plan in some detail before presenting it to NASA decision makers, but in this case, the craft had come a bit before the hour.

So, to see if the plan he had convinced NASA could be done really could be, he followed his intuition and experience and spent weeks in a long trial-and-error process, trying out different com-

binations of looping orbits to try to get from here to there. The basic strategy was to use the moon as a kind of cosmic slingshot, flying close by and picking up enough gravitational energy to propel the half-ton craft in the right direction for a rendezvous 44 million miles away.

"We get a kick in the pants from the moon's gravity," Goddard scientist Stephen Maran explained last week. "The spacecraft picks up speed from the fall towards the moon just like a skier picks up speed from gravity as he goes downhill."

Farquhar knew the orbit the spacecraft was in, and he knew the path it had to end up on to reach the comet. The problem was in getting from one to the other — getting the path to line up. It's a bit like knowing the way from your house out to a major road, and having directions from another main road to a friend's house. In order to drive over to visit your friend, you still have to figure out how to get from one main road to the other — and preferably by the most direct route.

The lowest point in Farquhar's personal orbit came when, after months of attempts, his assistant, David Dunham, brought out the printouts from his latest computer run — one that absolutely had to work or they would have been "back to square one" — and the orbits still stubbornly refused to line up. It was beginning to look like there might be the equivalent of an uncrossable river between one "road" and the other.

"I couldn't believe it," Farquhar says. "I mean, I figured it had to work. I remember driving home that night and I was totally depressed. I thought, 'son of a bitch, this thing is not going to work.' After all this — I've told all these people, I've got everybody cranked up on the administrator on down, saying we can do this thing. What have I done here? This time I've gone too far."

Homing in on answer

But this nadir was not to last long. "I got a call shortly after I got home. Dave Dunham said that

one case that he had run hadn't come back before, and he now had it, and then he told me the angle that it came in, and it overshot the angle that we had wanted. So, then I knew we had it bracketed. I knew there was no problem."

"Ultimately, we were going to come up with an answer. We didn't have it yet, but I was pretty confident from that moment on."

The answer, sure enough, did emerge. It turned out to involve five separate loops around the moon, the last of which was to bring the spacecraft within a hair-raising 75 miles of the lunar surface. In a field where most maneuvers involve millions, or at least thousands, of miles, that tactic was the equivalent of threading a cosmic needle at several thousand miles per hour, and made many of the Goddard planners nervous. Farquhar, however, was as unruffled as ever.

The spacecraft shot past the moon in December 1983, right on target toward the comet. At that moment, it was rechristened as the International Cometary Explorer, or ICE.

Little energy needed

The entire remarkable mission was achieved just by using the tiny maneuvering thrusters to nudge the craft gently in just the right direction at just the right time. These thrusters propel the craft in space much as a released child's balloon flies around the room as its air escapes — and with not much more power.

"The whole concept for this very surprising project came directly from the genius of Dr. Farquhar," Soviet space scientist Roald Sagdeev said last Thursday, the day after the successful penetration of the comet produced a wealth of surprising scientific data. "The event of yesterday was a very good inspiration for all of us."

Farquhar's mind, the day after the ICE triumph, was already way back out in space and far into the future. He was thinking about plans for a new spacecraft to be sent on an even more ambitious mission — one that would take it in for a close inspection of the sun, and then fly close to three different comets on the way back to allow scientists to compare their structure.

But with his vision firmly on the future, he could also now admit to his secret fears about the mission just completed, which had achieved such success, providing information that the scientists described as "fantastic," "a very rich base of data," "very exciting," and "a jewel of very high quality."

Farquhar said Thursday that "I thought we might go through [the comet] and just not see anything." With obvious relief at the way things turned out, he added in a half-whisper: "I really thought it might be a total fiasco."

Appendix D

NASA/NASM Joint Declaration of Intent

JOINT DECLARATION OF INTENT

Whereas on September 11, 1985, the International Cometary Explorer (ICE) made the historic first visit of a spacecraft to a comet, and

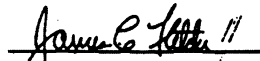
Whereas, the National Air and Space Museum (NASM) of the Smithsonian Institution has indicated its recognition of the historical significance of that comet encounter and of the ICE spacecraft, and

Whereas the ICE in its future orbital motions is expected to return to the vicinity of the Earth in August, 2014, when it might be retrievable, and

Whereas NASA has the authority to transfer vehicles to the NASM once their technical utility has been exhausted, and whereas NASM has the authority to accept such vehicles pursuant to combined authority of Section 503(c)(6) of the National Aeronautics and Space Act of 1958, as amended (42 U.S.C. 2423(c)(6)); Sections (4) and (8) of the National Air Museum Amendments Act of 1986 (20 U.S.C. 77(a), 77(d)); and the Agreement Between the National Aeronautics and Space Administration and the Smithsonian Institution Concerning the Custody and Management of NASA Historical Artifacts. (NMI 1052.85, March 14, 1987)

Now, therefore, on the occasion of the first anniversary of the encounter of the ICE with the Comet Giacobini-Zinner, NASA declares its intention to transfer title of the ICE spacecraft to the NASM upon such future date at which the technical utility of the spacecraft to NASA and other government agencies is exhausted.

And furthermore, NASM declares its intention to add the ICE spacecraft to the Smithsonian Institution's permanent collection upon NASA's aforementioned declaration and if the spacecraft is retrieved and returned to the surface of the Earth.


James C. Fletcher
Administrator
NASA

September 11, 1986


James C. Tyler
Acting Director
NASM

September 11, 1986

The last two paragraphs which may be obscured by the spacecraft in this reproduction state:

"Now, therefore, on the occasion of the first anniversary of the encounter of the ICE with the Comet Giacobini-Zinner, NASA declares its intention to transfer title of the ICE spacecraft to the NASM upon such future date at which the technical utility of the spacecraft to NASA and other government agencies is exhausted.

And furthermore, NASM declares its intention to add the ICE spacecraft to the Smithsonian Institution's permanent collection upon NASA's aforementioned declaration and if the spacecraft is retrieved and returned to the surface of the Earth."

Document D-1 Joint Declaration of Intent signed by Dr. James C. Fletcher for NASA
and Dr. James C. Tyler for NASM.

Appendix E

Memo Regarding Patent for Double Lunar-Swingby Trajectory Concept

August 29, 1979

TO: 580/Mission Support Computing and Analysis
Division/Dr. Robert W. Farquhar

FROM: 204/Patent Counsel

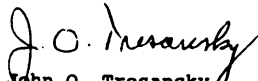
SUBJECT: Invention Disclosure Entitled; "Trajectory Concept
for Exploring the Earth's Geomagnetic Tail"

NASA Case No. GSC 12,556-1

An evaluation of your invention disclosure has recently taken place to aid me in recommending to NASA Headquarters whether patent action should be taken with respect to your invention. As a result of this evaluation, it has been recommended that your disclosure be placed in our inactive patent files. While it is recognized that the subject invention may have some patentable novelty, my action was based upon information as to technical significance of your invention and the ascertainment of the expected use by the Government thereof, which information was provided by personnel having supervisory technical cognizance of the subject area of the invention. Further, your invention disclosure was compared with other invention disclosures received by my Office to weigh the relative merits thereof, so that our time may be directed to the most valuable inventions from the standpoint of degree of novelty, past and expected future use by the United States and the probable success of obtaining worthwhile patent coverage thereon. The above evaluation indicated that there was insufficient reasons to warrant the expense of taking further patent action on your disclosure. However, if a significant governmental interest should develop in the future, it is requested that this Office be promptly notified so that the invention can, at that time, be reconsidered for further patent action.

You are advised that the disclosure of the subject invention may be published and disseminated through the NASA Technology Utilization Program.

In the event you desire to file an application at your own expense on this disclosure, it is requested that you inform this Office so that the necessary arrangements can be completed and a formal determination can be made to ascertain what rights the Government will obtain in any patent application based on this disclosure.


John O. Tresansky